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Dealna
A Secure University-Verified Marketplace

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Section – B

Title of Project: Dealna A Secure University - Verified Marketplace
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Key Areas: This project focuses on designing a university-based digital marketplace that combines item trading, user interaction, recommendation mechanisms, and integrated security controls to ensure trust, fairness, and system reliability.

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Abstract

Dealna is a secure mobile marketplace application designed exclusively for university communities, enabling students to buy, sell, and exchange items such as books, electronics, and daily student needs. The platform establishes a trusted environment through university email verification, role-based access control, and end-to-end encrypted messaging for secure communication. Users may choose to interact using either their real name or a nickname, providing privacy while maintaining accountability.

The system incorporates a content-based recommendation mechanism to suggest relevant listings based on user interests and browsing behavior, as well as a queue-based model that ensures fairness in Giveaway transactions. To enhance trust and reduce misuse, the platform applies moderation policies, rating mechanisms, and fraud-prevention controls. In addition to student users, Dealna supports verified external service providers through an administrative approval process, allowing trusted non-student services to participate in a controlled manner.

The application is designed as a university-scoped system, initially targeting **Birzeit University (BZU)** as a pilot deployment. Its architecture allows future expansion to multiple universities, each operating as an isolated and secure community. The project evaluation includes a comparative analysis with existing solutions, highlighting Dealna's advantages in security, trust, and university-specific access control.

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Chapter 1

Introduction

1.1 General overview

Dealna is a marketplace designed to solve real challenges faced by university communities when **buying and selling items**. Current student marketplaces rely heavily on social media groups, which suffer from clutter, lack of user verification, and weak security controls. **Dealna** introduces a structured and trusted environment built on university email authentication, encrypted communication, and intelligent mechanisms for recommendations and fraud detection.

The platform ensures fairness, trust, and an improved user experience through features such as queue-based reservation, Role - based access control, and anonymous posting. **Dealna** serves as a modern solution for campus trading, starting with **Birzeit University** and designed for future expansion across other universities.

1.2 Motivation

University students face several challenges when attempting to buy or sell used items on campus. Traditional platforms like social media groups are overcrowded with irrelevant posts and lack identity verification, allowing non students or scammers to infiltrate these communities. There are no proper filters to help users find what they need, no rating system to build trust, and no fair method for managing item reservations for Giveaways. As a result, students often experience confusion, wasted time, and security risks when using these platforms. There is a clear need for a secure, student focused marketplace that ensures verification, fairness, and trust while improving the overall experience of campus-based trading.

1.3 Method

1. **Frontend:** Flutter for mobile app.
2. **Backend:** A RESTful backend developed in Go, responsible for managing users, profiles, followers, items, chat functionality, and queue-based operations.
3. **Database:** PostgreSQL storing items, users, ratings, follow relationships, and images.

4. **Authentication:** University email verification with secure password storage (Salting + Hashing).

5. **Recommendation System:** Uses content-based filtering with TF-IDF and cosine similarity to suggest items related to a student's recent searches, viewed, or liked posts [1], [2], [3], [4].

6. **Queue & Hold System:** Items can be held by one user; others can join queue; automatic release after timeout.

7. **Chat:** Secure in-app messaging using SSL/TLS encryption during transmission and End-to-End Encryption (E2EE) to ensure that messages and attachments remain private and cannot be accessed by the server or any third party[5].

8. **Admin Panel:** Dashboard for managing items, users, flags, reports, and analytics.

9. **Hybrid Fraud Detection System:**

   - **Rule-Based Validation:** Enforces daily post limits, validates price ranges per category, and checks item completeness [6].
   - **Text-Based Duplicate Detection:** Uses TF-IDF vectorization combined with cosine similarity to identify duplicate or near-duplicate items, even when wording is slightly modified. items exceeding a predefined similarity threshold are automatically flagged [7], [8].
   - **Statistical Anomaly Detection:** Continuously calculates mean and standard deviation of prices within each category. items with prices deviating beyond three standard deviations are flagged as potential anomalies for admin review [9].
   - **Behavioral Analysis:** Monitors user posting frequency and patterns to detect spam or abnormal activity [10].
   - **Integration with RBAC:** Detected violations trigger dynamic access adjustments, from warnings to account downgrade (Verified Student → Limited Student) [11].

10. **Security & Access Control:**

    - **Session Timeout:** The system automatically logs the user out after a period of inactivity to prevent anyone else from using the account if the device is left unlocked, especially on shared devices[12].
    - **Single Active Session per User:** The system ensures that only one session is active per user. When logging in from a new device, the previous session is immediately terminated to enhance privacy and prevent account sharing[12].
    - **Dynamic Access Adjustment:** The system relies on monitoring verified complaints and user ratings. When violations are repeated, warnings are issued first, If the user receives repeated verified complaints or multiple low ratings, the admin may downgrade the account from Verified Student to Limited Student, where posting and trading permissions are revoked while basic browsing remains available.

- **Role-Based Access Control (RBAC):**The system relies on a role-based permissions model to define the tasks allowed for each category of users according to their role within the platform [11]:
  - **Admin:** Account management, reviewing ads and reports, handling reports, and general control from the admin panel.
  - **Verified Student:** Publishing ads, interacting with items, buying and selling, and messaging within the app.
  - **Limited Student :** Can browse only.
  - **Verified Providers:** Publishing their own services or offers only, without the ability to purchase or interact with other users' offers.
- **Secure Image Upload:** The system employs a secure image verification mechanism before storage, allowing only JPG ,JPEG and PNG files. The server verifies the true file type using Magic Numbers to prevent files disguised as image files. The image is then re-encoded internally to ensure the removal of any hidden or malicious content. Finally, the processed images are stored on a secure external storage service, while only the image URL is recorded in the database [13][14].
- **Input Validation:** All user input is validated and sanitized to prevent the injection of malicious content. The system treats all input strictly as data and never as executable code. Parameterized queries are enforced to ensure that user supplied input cannot alter the structure or logic of SQL statements, thereby mitigating SQL injection attacks[15].

1.4 Aim and Objectives

1. User Access

- Provide a trusted digital marketplace exclusive to verified students through university email authentication.
- Ensure secure password storage using hashing and salting.
- Support anonymous options to protect user identity when needed.

2. Item Discovery

- Simplify item discovery using advanced search filters (category, college, department, high-rated users).
- Recommend items based on user behaviour and recent interactions.

3. Trading Workflow

- Add a structured chat system to facilitate communication between users.
- Implement an item queue/hold mechanism to ensure fairness in Giveaways.

4. Admin Control

- Provide an admin web panel for content moderation and handling reports.
- Offer analytics and reporting tools to monitor platform performance.

5. Security

- Access is controlled using RBAC, ensuring that users are authorized based on their assigned roles.
- Enforce session management, such as limiting active sessions and automatic timeouts.
- Apply dynamic access adjustment when users violate rules or receive repeated complaints.
- Ensure secure image upload by verifying real file type .
- Prevent SQL injection attacks through strict input validation, treating all text as non-executable content.

6. Fraud Detection

- Limit daily posts and validate item completeness through rule-based checks.
- Detect duplicate items using TF-IDF text similarity analysis.
- Identify abnormal price patterns through statistical anomaly detection.
- Monitor user behaviour to detect unusual or suspicious activity.

1.5 Overview of the technical area

Dealna belongs to the technical field of electronic marketplace systems, combining concepts from software engineering and application security.

1.6 Overview of the report

Chapter 2 background. chapter 3 we will introduce the method.

Chapter 2

Background

This chapter provides an introduction to the proposed system by outlining its main ideas and context. Existing systems with similar objectives are reviewed and compared in order to highlight the gaps that motivated the development of our solution. In addition, the chapter presents the tools and technologies that will be used throughout the implementation of the system.

2.1 Details of relevant theory

2.1.1 Algorithms

1. **Recommendation and Fraud Detection Algorithms**

   - The system applies a unified TF-IDF with **cosine similarity** model to power both **recommendation** and **fraud detection**. Each item's title and description are transformed into TF-IDF vectors, emphasizing distinctive words while reducing the weight of common terms [1], [2].
   - **Cosine Similarity for Text Comparison:** Measures the angle between TF-IDF vectors to assess item similarity, making it scale-invariant and effective for both short and long descriptions. Huang (2008) demonstrated that cosine similarity outperforms Jaccard, Euclidean, and Manhattan distance for sparse, high-dimensional text data [3], [4].
   - For **recommendation**, items with high cosine similarity to a user's viewed or liked items are suggested (**personalized discovery**).
   - For **fraud detection**, items that exceed a predefined similarity threshold with existing ones are flagged as potential duplicates or spam. This approach ensures both **personalized discovery** and **content integrity**.
   - **Statistical Anomaly Detection:** Price anomalies are detected using statistical thresholds (mean $\pm$ 3 standard deviations per category). This method flags abnormally high or low prices that may indicate fraud or pricing errors, triggering automatic review [9].
   - **Behavioral Monitoring:** Tracks user posting frequency and activity patterns to identify spam or misuse. Detected violations are logged and used to trigger automated restrictions or moderation actions, ensuring system integrity and fair usage .

2. **Queue Management Algorithm:** FIFO logic with timeout .
3. **Moderation Rules:** Based on user behavior and rating trends.

2.2 Review of past/reported work

2.2.1 Related Work

There are many platforms and apps designed for university students that allow them to easily buy, sell, and exchange goods, such as UniExchange, Yupa, and Daleel Al Talib (an Arabic app) [16], [17], [18]. While Daleel Al Talib aims to facilitate exchanges between students, like other apps, it is limited to basic services like items, posting ads, and user communication. Most of these apps also lack a robust university verification system, advanced access control mechanisms, fairness in product exchange and reservation such as a queue/hold system, reliable user rating system, and secure messaging.

Dealna is the first Arabic application created especially for university students. It offers a closed and secure environment that relies on university email verification to make sure every user is a real student. For the current stage, Dealna will focus only on Birzeit University (BZU), providing a safe space for students to buy, sell, or exchange items within their own verified community. In the future, the app can grow to include other universities, where each one will have its own separate and secure community.

The application includes role-based access control to manage user permissions and a reservation and waiting list feature that help keep exchanges fair for everyone. Students can browse items easily using filters, post and manage their ads, and the app includes a rating system based on user feedback after each transaction to ensure a reliable reputation that fosters trust within the university community.

To maintain security, the app includes a dynamic mechanism that automatically adjusts user permissions depending on the number of complaints they receive. In addition, the system enforces secure input validation to prevent injection attacks, applies session management controls such as session timeouts and single active sessions.

Another key feature is the option to assign trusted users with special permissions through the Admin Panel. These users can manage certain types of posts, like those about university housing, to increase reliability and safety within the platform.

Dealna also uses an intelligent recommendation system that suggests suitable items for each student based on what they have previously searched for or chosen, making the experience more personalized and efficient [1], [2], [3], [4].

Finally, Dealna ensures privacy through robust conversation protection based on SSL/TLS encryption in transit and end-to-end encryption (E2EE), keeping all messages and attachments confidential and guaranteeing that no third party, including the server, can access their content. This level of protection is not mentioned in similar applications.

After comparing the features of the other applications mentioned with Dealna, it be-

comes clear that most rely on the basic functions of any student exchange system, but lack the simple yet crucial features that Dealna provides. These other applications lack robust verification systems and advanced mechanisms for access control and fair exchange. Dealna offers these features, in addition to others that provide a comprehensive, secure, and student-friendly environment, covering all student needs. Dealna offers an integrated model that combines university security and verification to ensure all users are genuine students, a role-based access control mechanism that dynamically adjusts user permissions based on system rules and user behavior, and a reservation and queuing system to guarantee fair exchanges. The user rating system after exchanges also fosters trust and credibility among community members, while encrypted messaging mechanisms using SSL/TLS and end-to-end encryption (E2EE) ensure privacy and security in communication, preventing the server or any third party from accessing or modifying message content. All these features combined make Dealna more than just a platform for student exchange; it's a comprehensive, secure, and customized solution for a closed university environment, capable of bridging the gaps in similar applications and providing a more reliable and effective experience for university students.

While some applications implement basic rule-based checks to limit posts and flag unusual prices [6], [10], they lack the depth and integration needed for comprehensive fraud prevention. Dealna goes beyond these basic thresholds by combining multiple detection layers: rule-based validation, TF-IDF text similarity analysis with cosine similarity for duplicate detection [7], [8], and statistical anomaly detection using mean and standard deviation to identify abnormal pricing patterns [9]. Furthermore, Dealna integrates these fraud detection mechanisms directly with its role-based access control and session management, creating a cohesive security framework where detected violations can dynamically adjust user permissions and access rights. This multi-layered, integrated approach sets Dealna apart from existing platforms that treat fraud detection as an isolated feature [6], [8], [10].

2.2.2 Applications Review

UniExchange Application

UniExchange is an application designed specifically for university students, functioning as a closed on-campus marketplace where students can buy, sell, and exchange products within their university community. The platform requires university email verification to ensure that all users are real students and includes a rating system to enhance trust after each transaction. It enforces fair pricing, typically requiring listed items to be priced between 50–60% of their original value to maintain fairness among students. All exchanges occur locally within the university, and users can report problematic buyers, sellers, or messages to maintain a safe environment. Overall, UniExchange offers a simple and convenient student-to-student exchange experience, with its core features focused on verification, basic safety, and ease of use.[17]

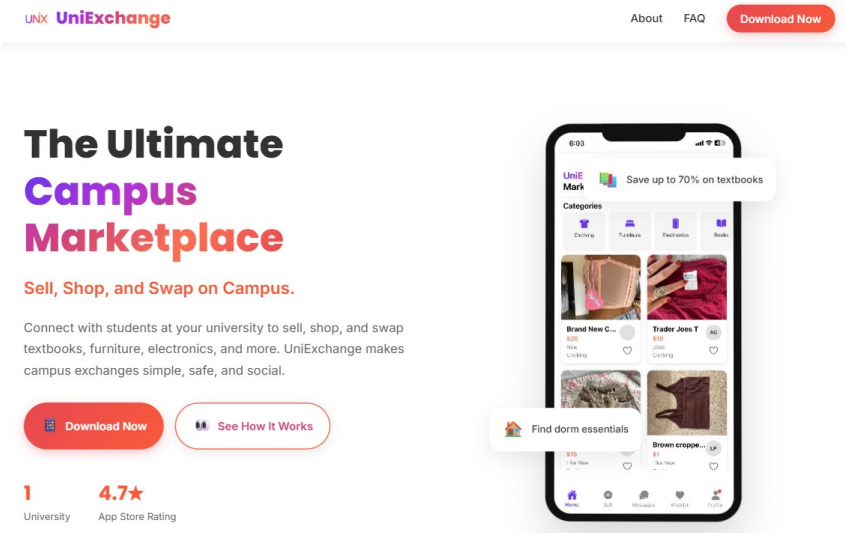

Figure 2.1: UniExchange Application [17].

YUPA Application

YUPA is an application aimed at university students, offering a service-based marketplace where students can provide tasks such as delivery, laundry, tutoring, photography, and other small services. The app includes several supportive features such as a profile builder, ranking and review systems, job notifications, dynamic in-app chat, background templates, and an integrated payment and payout system. It also provides a location changer tool that allows students to switch their service availability across different campuses. Based on user reviews, YUPA is generally considered reliable , limitations have been reported in post transaction communication, as messaging is automatically closed after job completion. Users have also noted minor usability challenges in browsing available jobs. According to its App Store data policy, YUPA collects different types of user information, including photos, approximate location, search history, names, email addresses, and in app messages, and may share certain details such as phone numbers or addresses depending on its data-handling policies. While YUPA offers a flexible and convenient environment for student services, it does not use university email verification and operates as an open marketplace rather than a closed, security restricted student platform.[16]

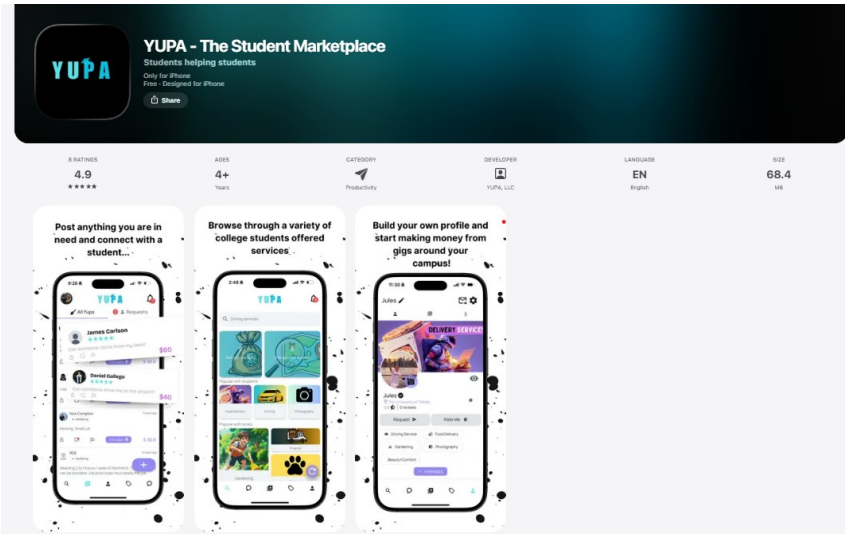

Figure 2.2: YUPA Application [16].

Daleel-AlTalib Application

Daleel Al Talib is a broad, general-purpose Arabic application that provides a wide range of services for students and the public. It functions similarly to comprehensive classifieds platforms such as “Haraj” , offering categories that include universities and institutes, visa services, real estate, transportation, books, electronics, furniture, vehicles, tourism, and various general services such as translation, maintenance, and delivery. The platform is not limited to students, as accounts may belong to companies, institutions, parents, and teachers. Unlike student-focused exchange applications, Daleel-AlTalib does not rely on university email verification and does not highlight advanced security features such as access control, duplicate post detection, or secure messaging. As a result, it operates as a wide public marketplace rather than a controlled, student exclusive exchange environment.[18]

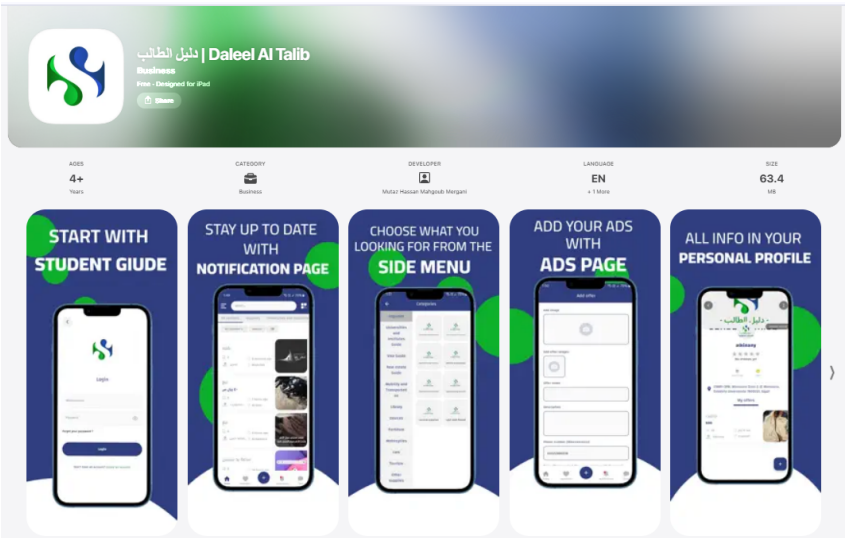

Figure 2.3: Daleel-AlTalib Application [18].

2.2.3 Detailed Comparison Between Apps

Feature	**UniExchange**	**YUPA**	**Daleel Al Talib**	**Dealna**
Target Audience	Verified university students	University students	Mixed public users (students +parents + companies)	Verified university students
University Email Verification	Required	Not specific	Not available	Required
Marketplace Type	Product exchange	Student services + basic items	Multi category classifieds	Product trading and exchange
Security Level	Basic	Basic	Low	Advanced (RBAC + E2EE)
Rating System	Available	Available	Not specific	Available
Queue / Hold System	Not available	Not available	Not available	Available
Fraud / Spam Detection	Basic reporting	Basic	Not specific	Multi-layer (rules + TF-IDF + anomaly detection)
Data Privacy	Minimal data	Collects and may share data	Not specified	High privacy (TLS + E2EE)
Trusted Providers Support	Not supported	Not supported	Not supported	Supported (verified + restricted)
Reporting System	User and message reporting	Reviews and issue reporting	Ads only reporting	User, and item reporting

Table 2.1: Detailed Comparison Between Apps table

2.2.4 Feature Descriptions

The table above compares different applications based on key functional and security features. The following descriptions clarify what each feature represents, to understands the purpose behind each comparison:

- **Target Audience:** Defines who the application is designed for, such as university students, the general public, or verified service providers.
- **University Email Verification:** A mechanism that ensures users belong to a specific university by requiring registration through an official university email domain.

- **Marketplace Type:** Specifies the nature of the platform whether it supports buying, selling, exchanging items, offering services, or a combination of these functions.
- **Security Level:** Defines how secure the application is, ranging from basic protections to advanced mechanisms like RBAC and end-to-end encryption used to safeguard data and prevent unauthorized access.
- **Rating System:** A feedback mechanism that allows users to evaluate each other after transactions to enhance trust and transparency.
- **Queue / Hold System:** A feature that organizes user requests fairly by placing them in a queue and serving users based on their position.
- **Fraud / Spam Detection:** Techniques and automated checks used to detect misleading content, duplicate posts, abnormal behavior, or other forms of misuse.
- **Data Privacy:** Indicates how user information is collected and protected. Apps with high privacy use encrypted communication and avoid unnecessary data collection, while apps with minimal or unspecified privacy may collect more data or provide weaker protection.
- **Trusted Providers Support:** A verification and approval process that allows external service providers (e.g., housing, pharmacies) to join the platform with additional review steps to ensure legitimacy.
- **Reporting System:** A mechanism that enables users to report inappropriate content or suspicious behavior so administrators can review and take appropriate action.

2.3 Tools and Technology

Development Environments

- **Visual Studio Code:** for Golang ,Flutter, Next, and documentation tasks.
- **Overleaf:** Collaborative platform for writing and managing the LaTeX report.

Programming Frameworks and Libraries

- **Golang:** REST API backend, security, authentication, and business logic.
- **Flutter (Dart):** Cross-platform mobile application development (Android/iOS).
- **Next.js** : Web interface for the Admin Panel.
- **Python:** Utility scripts for TF-IDF, anomaly detection, and data preprocessing.

Database and Storage

- **PostgreSQL:** Main relational database for users, items, queues, ratings, and permissions.

Testing and API Interaction

- **Postman:** Used for testing backend APIs and verifying request/response behavior.

Version Control and Collaboration

- **Git & GitHub:** Source control, issue tracking, and collaborative development.
- **Slack:** Team communication, coordination, and updates.

Diagrams and Design Tools

- **Draw.io:** For system architecture diagrams, UML diagrams , ERDs, and flowcharts.
- **Figma:** UI/UX design, mobile mockups, and collaborative prototyping.

Chapter 3

System Analysis and Design

This chapter presents the system analysis and design of the *Dealna* platform. It describes the overall system model, architectural structure, and key design decisions that guided the development process. The chapter focuses on how the system components interact, how responsibilities are separated between different layers, and how security and maintainability are achieved.

3.1 Product Description

Dealna is a university-oriented marketplace platform designed to enable safe and regulated trading among students and verified service providers. The system allows users to list items, participate in giveaways, complete transactions, communicate securely, and submit reports, all within a controlled academic environment. User authenticity is ensured through university email verification and role-based access control.

The platform supports multiple user roles, including verified students, limited students, verified service providers, and administrators. Each role is associated with a specific set of permissions that determines accessible features. Administrative users interact with the system through a dedicated web-based admin panel, while students and providers primarily use the mobile application.

3.1.1 System Objectives

The main objectives of the Dealna system are:

- Provide a secure and trusted marketplace tailored for university students.
- Enforce user authenticity through university email verification and controlled role assignment.
- Maintain a clear separation between frontend presentation layers and backend business logic.
- Reduce fraudulent behavior through transaction validation, rating stability mechanisms, and administrative oversight.
- Support common marketplace activities such as item listing, giveaways, transactions, and communication.

- Enable efficient moderation and management through a dedicated administrative interface.
- Design the system to allow future expansion to additional universities.

3.1.2 System Main Features

The Dealna platform provides the following core features:

- User registration and authentication using verified university email addresses.
- Role-based access control distinguishing between verified students, limited students, verified service providers, and administrators.
- Item listing creation, editing, and removal with image attachments and categories.
- Giveaway functionality implemented using a queue-based mechanism.
- Transaction management with clearly defined states.
- Rating and review system with fraud-aware stabilization mechanisms.
- Automated fraud detection with administrative review support.
- Trusted service provider support through an approval workflow.
- Secure one-to-one encrypted messaging.
- Reporting and moderation tools.
- Event-based notification system.
- Administrative management through a Next.js-based admin panel.

3.1.3 Operating Environments

The Dealna system operates in the following environments:

- **Backend:** Golang REST APIs running on a server environment.
- **Mobile Client:** Flutter application for Android and iOS.
- **Admin Panel:** Web-based interface built with Next.js.
- **Database:** Relational database ensuring transactional consistency.
- **User Environment:** Mobile devices with internet connectivity.

3.1.4 Constraints

The system design is subject to the following constraints:

- Academic time and scope limitations.
- Security requirements for authentication and authorization.
- Technology constraints imposed by chosen frameworks.
- Initial deployment limited to a single university.
- Limited infrastructure and testing resources.

3.1.5 User Requirements

UR-1 Secure Registration and Login
Users need to be able to register and log in securely using their university email (for students) or business email (for providers), with protection against weak passwords and abuse.

UR-2 University-Scoped Community
Students need a marketplace limited to their own university community so they only interact with trusted, verified students.

UR-3 Privacy and Anonymity
Users need to keep their real identity (email, academic data) private and use a public nickname when interacting on items and chats.

UR-4 Easy Items Creation and Management
Users need to create, edit, and delete Items with images, prices, categories, and conditions in a simple way.

UR-5 Search and Discovery
Users need to search for items with free-text queries and filters and see relevant results based on the text they enter.

UR-6 Safe Communication
Users need a secure chat to discuss items without exposing their contact details, and they expect chats to be private and encrypted.

UR-7 Fair Giveaway Queue
Users need a fair way to participate in giveaways, with a clear queue, turn notifications, and timeouts if someone does not respond.

UR-8 Clear Transaction Status
Users need to know whether an item is available, reserved, or completed, and to confirm when a transaction is finished.

UR-9 Notifications
Users need to receive notifications for important events such as queue turns, reservation decisions, completion requests, and account status changes.

UR-10 Safety and Reporting
Users need to be able to report suspicious users, fraudulent items, or inappropriate content and expect the system to react.

UR-11 Verified Providers
Users need a way to distinguish trusted external service providers (e.g., housing, pharmacy, printing) from regular student users.

UR-12 Fraud Protection
Users need protection from scams and fraudulent behavior through automated fraud detection and admin monitoring.

UR-13 Recommendations
Users need the system to help them discover relevant items by recommending items related to what they search for.

3.1.6 System Requirements

Functional Requirements (FR)

1. User Registration

FR-1.1 Account Type Selection
The system shall allow users to choose their account type during registration:

- Student Account
- Verified Provider Account

FR-1.2 Student Registration Input
The system shall allow students to register by providing:

- Display Name (Real name or Nickname - user chooses freely)
- University Email
- Password
- Major (optional)
- Academic year (optional)

FR-1.3 University Email Domain Validation
The system shall validate that the provided email belongs to an approved university domain.

FR-1.4 Email Activation Link Verification
The system shall:

- Send a unique account activation link to the user's university email.
- Allow registration to continue only after the activation link is opened.
- Ensure that each activation link is single-use and becomes invalid immediately after successful activation.

FR-1.5 Incorrect Activation Code Handling
The system shall:

- Allow the user to request the activation link again if it expires
- Limit activation link resend attempts to 3 times within 30 minutes.
- Temporarily block further resend attempts for 30 minutes after exceeding the limit.

2. Provider Registration and Management

FR-2.1 Email Submission
The system shall allow external service providers to begin the registration process by entering their email address only.

- The system shall verify that the email is not already associated with an existing provider account.
- If valid and unique, the system shall send an activation link to the provided email.

FR-2.2 Email Activation Link Verification
The system shall send an activation link to the provider's email.

- The provider must click the link to verify ownership.
- Until the link is activated, the account state shall be: Awaiting Email Verification.
- After activation, the provider shall be redirected to the Complete Registration Form.

FR-2.3 Complete Registration Form
After email activation, the system shall prompt the provider to complete their registration by submitting the following information:
Non-Optional:

- Business / Provider Name
- Contact Phone Number
- Business Type such as housing, pharmacy, print shop
- Business Location / Address
- Password
- Legal Documents / Proof of Service (official documents proving the provider is authorized to offer the service)

Optional:

- Additional supporting documents (certificates, portfolio, etc.)
- The system shall prevent duplicate accounts based on email and phone number.
- After submission, the account state shall become: Pending Admin Approval.

FR-2.4 Admin Review and Decision
The system shall allow administrators to review provider applications and take one of the following actions:
Approve:

- Upgrade the account to Verified Provider
- Send approval notification email with guidelines

- Display the Verified Provider badge on the provider's profile and items
- Enable posting access only in provider-designated categories

Reject:

- Deny the application
- Send the rejection reason to the provider
- Allow the provider to submit a new application if desired

Request Additional Information:

- Request missing or unclear documents
- Change account state to: Pending – Additional Info Required
- After resubmission, return the account to: Pending Admin Approval

FR-2.5 Unique Account Constraint
The system shall prevent duplicate provider accounts based on email and phone number.
FR-2.6 Provider Dashboard
The provider shall be able to track their application status in the dashboard, showing:

- Status: Awaiting Email Verification / Email Verified – Complete Registration / Pending Admin Approval / Pending – Additional Info / Rejected / Verified Provider
- Submitted document
- Notifications from admin (approval, rejection, or information request)

FR-2.7 Role Assignment After Registration
The system shall automatically assign one of the following roles:

- Verified Student
- Verified Provider (after admin approval)

3. Login & Session Management

FR-3.1 Login Input
The system shall allow users to sign in using:

- Email
- Password

FR-3.2 Account Status Verification During login, the system shall verify:

- Account is activated
- Provider is approved (if applicable)
- Account is not suspended or restricted

FR-3.3 Secure Login
The system shall authenticate users using email and password.
FR-3.4 Password Hashing
Passwords shall be stored in hashed and encrypted form.
FR-3.5 Failed Login Attempts
The system shall temporarily restrict login attempts after multiple consecutive failed authentication attempts.
FR-3.6 Token-Based Sessions (JWT)
The system shall use secure tokens (JWT) for session management.
FR-3.7 Single Active Session Enforcement
The system shall allow only one active session per user. Logging in from a new device shall invalidate previous sessions.
FR-3.8 Session Expiration
Inactive sessions shall automatically expire after a predefined period of inactivity.
FR-3.9 Session Invalidation on Logout:
The system shall invalidate the active session token immediately upon user logout to prevent further use.
FR-3.10 Forgot Password – Activation Code Reset
The system shall allow users to reset their password by sending an activation code to their registered email or phone number. The user must enter the code before creating a new password.

4. Multi-University Support

FR-4.1 University Identification
The system shall detect and assign a user's university based on their email domain.
FR-4.2 University Table
The system shall maintain a university table containing:

- university_id
- name
- domain
- logo
- status (active/inactive)

FR-4.3 Scalable Expansion (Open–Closed Principle)
Adding new universities shall require only inserting a new row in the University table, without modifying core logic.
FR-4.4 University-Based Isolation
Users shall interact only with items, chats, and communities of their own university.

5. Public vs Private Identity

FR-5.1 Private Identity
The system shall store but never publicly show:

- University email
- Major
- Academic year

FR-5.2 Public Identity
The system shall show publicly only:

- Display name
- Profile picture
- Rating

FR-5.3 Anonymity Support
The system shall allow the user to choose between showing their real name or a nickname (user-chosen or system-generated). This public identity shall appear on items, chats, and interactions.
FR-5.4 Display Name Change Rate Limitation
The system shall limit display name changes to once every 30 days.

6. Display Name Management

FR-6.1 Unique Display Name Constraint
Display names shall be globally unique using a database unique constraint.
FR-6.2 System-Generated Nicknames
The system shall provide an option for automatically generating a safe, unique nickname.
FR-6.3 Profanity Filtering
Nickname input shall be filtered to reject offensive or inappropriate words.
FR-6.4 Display Name Editing
Users may edit their display name, subject to:

- Uniqueness check
- Profanity filtering
- Rate limit (once every 30 days)

7. Profile & Account Management

FR-7.1 Profile Editing
Users shall be able to:

- Update personal details
- Change profile picture
- Configure notification preferences

8. Rating System

FR-8.1 Rating Availability
Users shall only be able to rate another user after a transaction is completed and both parties have confirmed completion (sale or exchange).
FR-8.2 Rating Scale
The system shall use a 5-star rating scale where:

- 1 star represents a very poor experience.
- 5 stars represent an excellent experience.

FR-8.3 Rating Calculation
A user's public rating shall be calculated using a Bayesian average to ensure fairness and rating stability. *For detailed explanation and algorithm comparison, click here.*
FR-8.4 Rating Integrity Protection
The system shall:

- Prevent repeated reciprocal ratings between the same users within a short time period.
- Detect abnormal rating behavior patterns.
- Flag suspicious rating activity for administrative review.

FR-8.5 Public Display of Ratings
User ratings shall be publicly visible, showing:

- Average star rating.
- Total number of ratings.

FR-8.6 Rating Impact on Account Privileges
The system shall dynamically adjust user privileges, including daily posting limits, based on the user's rating:

- Users with a rating of **4.5 or higher** may create up to **10 listings per day**.
- Users with a rating between **3.0 and 4.49** may create up to **7 listings per day**.
- Users with a rating below **3.0** shall be restricted to the base posting limit of **4 listings per day**.

FR-8.7 Rating Transparency
Ratings shall:

- Be anonymous to other users.
- Remain accessible to administrators for moderation and dispute handling.

9. items Management

FR-9.1 Create item
Users shall be able to create items with:

- Item title
- Description
- Category
- Price (optional for exchange items)
- Condition (new/used)
- Images (1–5)
- Pickup location

FR-9.2 Edit/Delete items
Users shall be able to edit or delete their active items at any time.

10. Search & Discovery

FR-10.1 Search Engine
The system shall allow users to search items using free-text queries that:

- Match phrases inside titles and descriptions
- Support multi-word queries

FR-10.2 Filters
Users shall filter items by:

- Category
- Condition
- Price
- Most recent
- Items posted by high-rated users

11. Chat & Communication

FR-11.1 Direct Messaging
The system shall support real-time messaging. Users may:

- Send/receive encrypted text messages
- Send/receive encrypted images and attachments
- View message status (sent, delivered, read)
- Access chat history

FR-11.2 Secure Messaging (E2EE)
The system shall apply end-to-end encryption by encrypting all messages and attachments on the sender's device using a shared chat-specific encryption key. The system shall ensure that:

- Text and images are encrypted client-side
- The server cannot read message content
- Only minimal metadata is stored
- Each chat has its own encryption key

FR-11.3 Chat Access Control

- Regular users may send/receive chats
- Admins may access metadata only
- Suspended/banned users lose chat access immediately

12. Queue System (Giveaway Only)

FR-12.1 Queue Enrollment
For items marked as "Giveaway," the system shall allow users to:

- Join a queue specific to the giveaway item.
- View their current position in the queue.
- Receive a notification when they become the first in the queue.

FR-12.2 Reservation Request (First Position Only)
The system shall enforce that:

- Only the user who is first in the queue may send a **reservation request** to the item owner.
- No other user may send a reservation request until they reach first position.

FR-12.3 Reservation Flow
When a user is the first in the queue:

- They shall receive a "Your Turn" notification.
- They must send a reservation request within the allowed time.
- If the owner receives a reservation request, they shall either:
  - Accept the reservation request
  - Reject the request (the user remains first until the turn expires or the owner rejects).

FR-12.4 Turn Expiration Handling
The system shall:

- Automatically remove the first user from the queue if they fail to send a reservation request within the allowed time.
- Promote the next user to the first position.
- Notify the newly promoted user that their turn has begun.
- Return the item to “Available” if the queue becomes empty.

FR-12.5 Item Owner Actions
For giveaway items, item owners shall be able to:

- Accept or reject incoming reservation requests.
- Remove the item from the marketplace, which cancels the entire queue and notifies all users in the queue that the item is no longer available.

FR-12.6 Sale items (Non-Giveaway)
For regular sale items:

- No queue shall be used.
- Buyers may contact the seller directly through chat.
- The seller may freely choose whom to sell to.

13. Transactions & Status Tracking

FR-13.1 Transaction States
Each item shall have one of the following statuses:

- Available
- Reserved
- Completed

FR-13.2 Completion Flow
A transaction shall be completed when:

- Both users confirm
- The item is archived

14. Notifications System (Sales Only)

Sales Completion Flow (No Queue - Direct)
FR-14.1 Multiple buyers may send purchase requests for the same item.
FR-14.2 All requests shall remain in a pending state until the seller accepts one buyer.
FR-14.3 Once the seller accepts a buyer, all other requests shall be frozen and cannot be approved.
FR-14.4 Completion must be initiated by the seller, who sends a "Completion Request" to the buyer.
FR-14.5 Both users must confirm the completion for the deal to finalize.
FR-14.6 When the deal is completed:

- The item is archived.
- All pending buyers are rejected and notified: “Item X has been sold.
- A rating request is triggered for the buyer only.

FR-14.7 If the seller or buyer cancels the accepted reservation:

- Previous pending requests become active again.
- Buyer/seller will be notified for that cancellation.
- Seller may choose another pending request.

15. Notifications System (Giveaway Only)

FR-15.1 Turn Notification
The system shall notify the next user in the queue when it becomes their turn and they are eligible to send a reservation request.
FR-15.2 Turn Timeout Notification
The system shall notify a user when their turn expires because they failed to send a reservation request within the allowed time.
FR-15.3 Reservation Request Notification
The system shall notify the Item Owner when the first user in the queue sends a reservation request.
FR-15.4 Reservation Acceptance Notification
The system shall notify the requesting user when the Item Owner accepts their reservation request.
FR-15.5 Reservation Rejection Notification
The system shall notify the requesting user when the Item Owner rejects their reservation request.
FR-15.6 Reservation Rejection Effects
When a reservation request is rejected:

- The requesting user is removed from the first position.
- The next user in the queue becomes first.
- The next user receives a “Your turn” notification.

FR-15.7 Reservation Cancellation
If the approved receiver cancels their reservation:

- The Item Owner is notified.
- The receiver is removed from the queue.
- The next user in the queue is notified that it is now their turn.

FR-15.8 Reservation Cancellation
If the Item Owner cancels the accepted reservation:

- The receiver is notified.

- All other users retain their queue positions.
- The next user in the queue is notified that it is now their turn.

FR-15.9 Completion Request Notification
The system shall notify the receiver when the Item Owner sends a "Confirm Giveaway Completion" request.
FR-15.10 Completion Confirmation Notification
Once the receiver confirms the completion request:

- The Item Owner is notified.
- The item is archived.
- All users in the queue are notified that the item has been given away.

16. Admin Dashboard

16.1 User Management
16.1.1 View & Search Users
Admins shall view:

- Display Name
- University email
- User role
- Rating
- Account status

Admins shall search by:

- Display Name
- Email

16.1.2 Account Moderation
Admins may:

- Suspend users
- Ban users
- View rating trends
- View complaint history
- Restore suspended accounts

16.1.3 Verified Provider Requests
Admins may:

- View pending provider requests

- Review submitted documents
- Approve or reject requests
- Revoke status at any time

16.2 items Management
Admins may:

- View all items with filters (category, price range, posting date, reported items)
- Delete inappropriate items
- View item details
- Manage reports on items

16.3 Analytics Dashboard
16.3.1 User Analytics
Shows:

- Top-rated users
- Lowest-rated users
- Active vs. suspended users count

16.3.2 items Analytics
Shows:

- Total items per category
- Deleted or reported items count

16.4 Reports & Flags
Admins may:

- View all user reports
- View report details (reporter, accused user/item, timestamp, description)
- Mark reports as resolved, under investigation, or escalated
- Apply sanctions (warning, suspension, ban)

17. Reporting & Safety

FR-17.1 User Reporting
Users shall be able to report:

- items (fraud, misinformation, inappropriate content, duplicates)
- Users (harassment, scam attempts, suspicious behavior)
- Attach optional screenshots

FR-17.2 Report Submission Details
A report shall include:

- Reporting user ID
- Reported user/item ID
- Reason + optional text
- Timestamp
- Optional attachments

18. Fraud Detection

FR-18.1 Duplicate Listing Detection (TF-IDF)
The system shall apply TF-IDF vectorization with cosine similarity to detect duplicate or highly similar listings.
TF-IDF shall be computed using the listing's title and description combined .
Listings that exceed a predefined similarity threshold shall be automatically flagged for admin review.
FR-18.2 Daily Posting Limits
The system shall enforce a maximum number of listings a user may post per day. Users who reach the limit shall be temporarily restricted from creating additional posts.
FR-18.3 Dynamic Posting Limits (Based on Rating)
The system shall dynamically adjust daily posting limits according to the user's rating:

- Users with a rating of **4.5 or higher** may create up to **10 listings per day**.
- Users with a rating between **3.0 and 4.49** may create up to **7 listings per day**.
- Users with a rating of **2.0 or below** shall be restricted to the base posting limit of **4 listings per day**.

FR-18.4 Transaction Gaming Prevention
The system shall prevent two users from repeatedly creating and completing transactions solely to inflate their ratings.
Suspicious rating patterns shall trigger warnings, temporary rating freeze, or admin review.*For a detailed explanation of the detection mechanisms and enforcement logic, click here.*
FR-18.5 Multi-Account Abuse Prevention
The system shall detect and flag potential multi-account abuse by monitoring:

- repeated interactions between the same two accounts
- repeated posting of similar items
- abnormal activity patterns

FR-18.6 Flagged Listings Handling
Any listing flagged by the fraud-detection rules shall be:

- temporarily hidden or marked as "Under Review"

- forwarded to administrators for manual inspection

FR-18.7 Repeated Violations
Users who repeatedly trigger fraud-detection rules shall face progressive penalties:

- warning
- posting limitation
- temporary suspension
- permanent ban (in severe or continuous misuse)

FR-18.8 Integration With Rating System
If a transaction is flagged as fraudulent, the system shall:

- freeze the associated ratings
- exclude those ratings from the user's average

19. Recommendation System

FR-19.1 Basis of Recommendation
The recommendation engine shall function exclusively based on:

- User search text input
- Textual similarity between search input and active item titles and descriptions

FR-19.2 TF-IDF Text Vectorization
The system shall convert all item titles, item descriptions, and user search queries into TF-IDF vectors to:

- Treat distinctive item terms (e.g., "MacBook Air M1", "IPhone 14 Pro") as high-weight similarity features
- Automatically reduce the influence of frequently occurring marketplace terms through inverse frequency weighting
- Maintain a continuously updated TF-IDF index of all active items

FR-19.3 Cosine Similarity Ranking
For every search action, the system shall:

- Compute cosine similarity scores between the search query vector and each active item title and description vector
- Rank all matching items strictly from highest similarity score to lowest
- Display the highest-scoring items at the top of the recommendation output

FR-19.4 Cold-Start Handling (No Search Performed)
When no search input exists:

- The system shall display a neutral discovery feed consisting of recently posted items

- No similarity-based ordering shall be applied
- Search activity shall immediately shift the feed into similarity-based ranking mode

FR-19.5 Real-Time Recommendation Update
The recommendation list shall update immediately when:

- A user submits new search keywords
- A new item is posted that textually matches current search input vectors

FR-19.6 High-Volume Search Handling
If the user performs multiple searches:

- Only the most recent five (5) search inputs shall be retained for similarity ranking
- Older search inputs shall be excluded to maintain relevance and prevent noise

Non-Functional Requirements (NFR)

NFR-1 Media Constraints

- Maximum allowed image size per upload shall not exceed 5 MB.
- Supported image formats: JPG, PNG ,JPEG.
- Maximum number of images per item shall be 5.

NFR-2 Posting Constraints

- The system shall limit users to a maximum of 5 active items per day (configurable by admin).
- Duplicate posting detection (TF-IDF) must run automatically on each new item.

NFR-3 Queue / Hold Constraints

- Queue reservation timeout shall be 1 hour .
- If timeout expires, the system shall automatically move to the next user in the queue.
- items shall automatically freeze when in Reserved state.

NFR-4 Verification Mechanism Constraints

- Account activation links or activation codes shall be single use and expire after 30 minutes .
- OTP codes shall expire within 10 minutes.
- OTP verification attempts shall be limited to 3 consecutive tries, after which the system shall temporarily block further attempts.

NFR-5 Session Security Constraints

- Session inactivity timeout shall be set to 25 minutes, after which the user is automatically logged out.
- Only one active session is allowed per user at any given time.
- A new session token is generated for each successful login.
- Session tokens become invalid immediately after user logout.

NFR-6 Password Policy Constraints

- Minimum length: 8 characters.
- Mixed uppercase and lowercase letters
- At least one number and one symbol

NFR-7 Nickname / Display Name Constraints

- Display name changes shall have a rate limit of once every 30 days.
- System-generated nicknames must always be unique.

NFR-8 Verified Provider Constraints

- Verified Providers can only post in specific categories approved by admins (e.g., housing, pharmacy items).
- Providers must submit documentation before approval.

NFR-9 Data Storage Limits

- Encrypted chat messages may be retained without a predefined time limit, while allowing users to manually delete their chat history at any time.
- The maximum allowed size for report attachments shall be limited to 3 MB per file to reduce storage abuse and ensure efficient system performance. [19]

3.1.7 Security Requirment

1. Identify Assets

- **User Accounts and Passwords:** This includes login information such as email addresses and passwords, as well as account status such as active or inactive. It must be protected to prevent unauthorized access to accounts.
- **Personal Information :** This include Information that users share about themselves, such as their university email address or other details that system as optional require. keep this information private prevents its misuse.

- **Roles and Permissions :** This include the information define each person's permissions within the system. If it's not secured, some individuals may use features they are not authorized to use.
- **Listings Data :** All details of items or ads posted on the platform. Altering this information without permission may compromise its accuracy and reduce user trust.
- **Transactions :** This include Information about past transaction , purchases, and activities between users. This information is central to the system .
- **Messages and Communication :** This include Messages exchanged by users. These often contain private details about deals or transactions and must be protected.
- **Ratings and Reputation:** This include user ratings . Change this data can lead to unfair outcomes and damage trust.
- **Reporting and Modertion :** Reports submitted by users and actions taken by administrators. This helps protect the Application security and integrity for everyone.
- **System Logs :** This include logs of important actions such as logins, transactions, and administrator activity. These logs are useful for tracking problems and resolving security vulnerabilities.
- **System Availability :** The system must operate easly and without interruptions so that users can rely on it at any time.

2. Identify Threats

- **Unauthorized Access :** someone could try to enter the application without having permission. If this happens, they might end up using another user's account or accessing features they are not authorized to use.
- **Data Modification :** system data such as listings, transactions, or ratings could be changed by unauthorized. This may cause incorrect information to appear and confuse users.
- **Personal Data Exposure :** Users share personal and academic details with the system. If this data is not handled carefully, it could be exposed or misused, which directly affects user privacy.
- **Fraudulent Activities :** Some users may try to exploit the platform by creating fake transactions or manipulating certain steps to benefit themselves.
- **Messaging Abuse:** the messaging feature is meant for communication, but it could be misused. For example, users might send spam messages .
- **System Unavailability:** There may be situations where the system becomes unavailable for a short period of time due to technical problems or heavy misuse, which prevents users from accessing services normally.

3. Identify Security Goal

- **SG1:**The system should ensure that only authorized users can access their accounts and use system features.
- **SG2:**The system should protect personal and academic user data and prevent it from being exposed to unauthorized parties.
- **SG3:**The system should make sure that users can only perform actions that are allowed based on their assigned roles.
- **SG4:**The system should maintain the accuracy and integrity of system data, such as listings, transactions, and ratings.
- **SG5:**The system should reduce the risk of fraud, fake transactions, and misuse of system features.
- **SG6:**The system should provide a safe communication environment and limit the misuse of the messaging feature.
- **SG7:**The system should remain available and minimize service interruptions caused by misuse or technical issues.

Security Requirements

Based on the identified assets, threats, and security goals, the following security requirements are defined to ensure the confidentiality, integrity, and availability of the system.

- **SR-1 Secure Authentication:** The system shall securely authenticate users using strong password protection mechanisms such as hashing to prevent unauthorized access.
- **SR-2 Access Control and Authorization:** The system shall enforce role-based access control to ensure that users can only perform actions permitted by their assigned roles.
- **SR-3 Data Privacy Protection:** The system shall protect personal and academic user data and prevent its exposure or misuse by unauthorized parties.
- **SR-4 Secure Communication:** The system shall protect user messages and communications using encrypted channels to prevent eavesdropping and data leakage.
- **SR-5 Secure File Upload Handling:** The system shall validate uploaded files to prevent the upload of malicious or harmful content.
- **SR-6 Data Integrity Protection:** The system shall ensure the integrity of system data such as listings, transactions, and ratings, and prevent unauthorized modification.
- **SR-7 Fraud Detection and Prevention:** The system shall detect and reduce fraudulent activities, such as fake transactions and manipulated ratings through monitoring and administrative controls.

- **SR-8 Reporting and Moderation:** The system shall provide a reporting mechanism that allows users to report suspicious behavior and enables administrators to take appropriate moderation actions.
- **SR-9 System Availability:** The system shall ensure service availability and minimize interruptions caused by misuse or technical issues.

3.2 Functional Decomposition

3.2.1 Actors

- **Verified Student:** A verified and registered university student whose university email address has been verified. The student can browse and search for products, create and manage product lists, join waiting queue, communicate with other users, rate users after transactions are completed, and submit reports.
- **Limited Student:** A previously verified and registered student account with limited privileges for a temporary period due to a violation of system rules. The user can browse and search for products but cannot post, create lists, exchange transactions, or send messages.
- **Verified Provider:** An external service provider verified and approved by the system administrator. The provider can publish their services in specific categories but cannot purchase products or interact as a user.
- **Admin:** The system administrator, responsible for managing users, reviewing reports, reviewing service provider requests, monitoring fraud alerts, and applying penalties.

3.2.2 Use Cases

UseCase-1: Login

Actor: Verified Student, Limited Student, Verified Provider, Admin.
Primary Action:

- Authenticate users using registered email and password.
- Verify account status and role permissions.
- Grant access to authorized system features.

UseCase-2: Browse & Search Items

Actor: Verified Student, Limited Student.
Primary Action:

- View available items without performing a search.
- Navigate between items based on categories or recent listings.
- Search for items using keyword based word.

UseCase-3: View Item Details

Actor: Verified Student, Limited Student, Admin.
Primary Action:

- View all available item images and description.
- View and evaluate a seller profile.
- check the queue position/status .

UseCase-4: Manage Listings

Actor: Verified Student , Verified Provider .
Primary Action:

- Create a new listing with its details.
- Upload images related to the listing.
- Update listing information when needed .
- Delete the listing if it is no longer available.
- Listings are published based on user role:
  - Verified students may post in general student categories.
  - Verified providers may post only in provider specific categories.

UseCase-5: Join Queue

Actor: Verified Student .
Primary Action:

- Join the waiting queue for the exchange of item.
- check position in the queue .
- receive a notification when it's user turn.

UseCase-6: Send/Receive Messages

Actor: Verified Student .
Primary Action:

- Send Messages to other user.
- Receive and read incoming messages.
- Communicate securely during transactions using encrypted messaging .

UseCase-7: Rate user

Actor: Verified Student .
Primary Action:

- After transaction , the user can rate the other user.
- User can also submit a feedback.

UseCase-8: Report Item/User

Actor: Verified Student .
Primary Action:

- User can report someone or something that abusing the system's conduct.
- User select report type such as fraud, duplicate item, scam, harassment, or inappropriate content.
- User can optionally upload file .

UseCase-9: Register as Provider

Actor: Verified Provider .
Primary Action:

- Submit initial form of provider register request .
- Provider verifies his identity by open an email activation links .
- Provider submit another form that contain required business and service verification document .
- Provider become in waiting status .

UseCase-10: Track Provider Application Status

Actor: Verified Provider .
Primary Action:

- View current application status such as pending, approved, rejected from dashboard .
- Provider receive email about admin request or decisions .
- Provider receive the admin feedback .

UseCase-11: Review provider Application

Actor: Admin .
Primary Action:

- Admin review submitted provider applications.
- Admin approval, reject, or request additional information .
- Update provider account status .

UseCase-12: Manage Users

Actor: Admin .
Primary Action:

- Admin view user accounts and roles .
- Admin restore user access after review .

UseCase-13: View Analysis dashboard

Actor: Admin .
Primary Action:

- Monitor user activity and marketplace trends.
- Monitor number of user that registration and system growth .

UseCase-14: Monitor Fraud Alerts

Actor: Admin .
Primary Action:

- Monitoring fraud and scam alerts originating from the system .
- Monitoring suspicious activity.
- Taking preventative measures to maintain system security.

UseCase-15: Investigate Reports

Actor: Admin .
Primary Action:

- Reviewing submitted user reports .
- Verify the accuracy of the report.
- Taking appropriate action if the report violates the terms of the system.

3.2.3 Use Cases Diagram

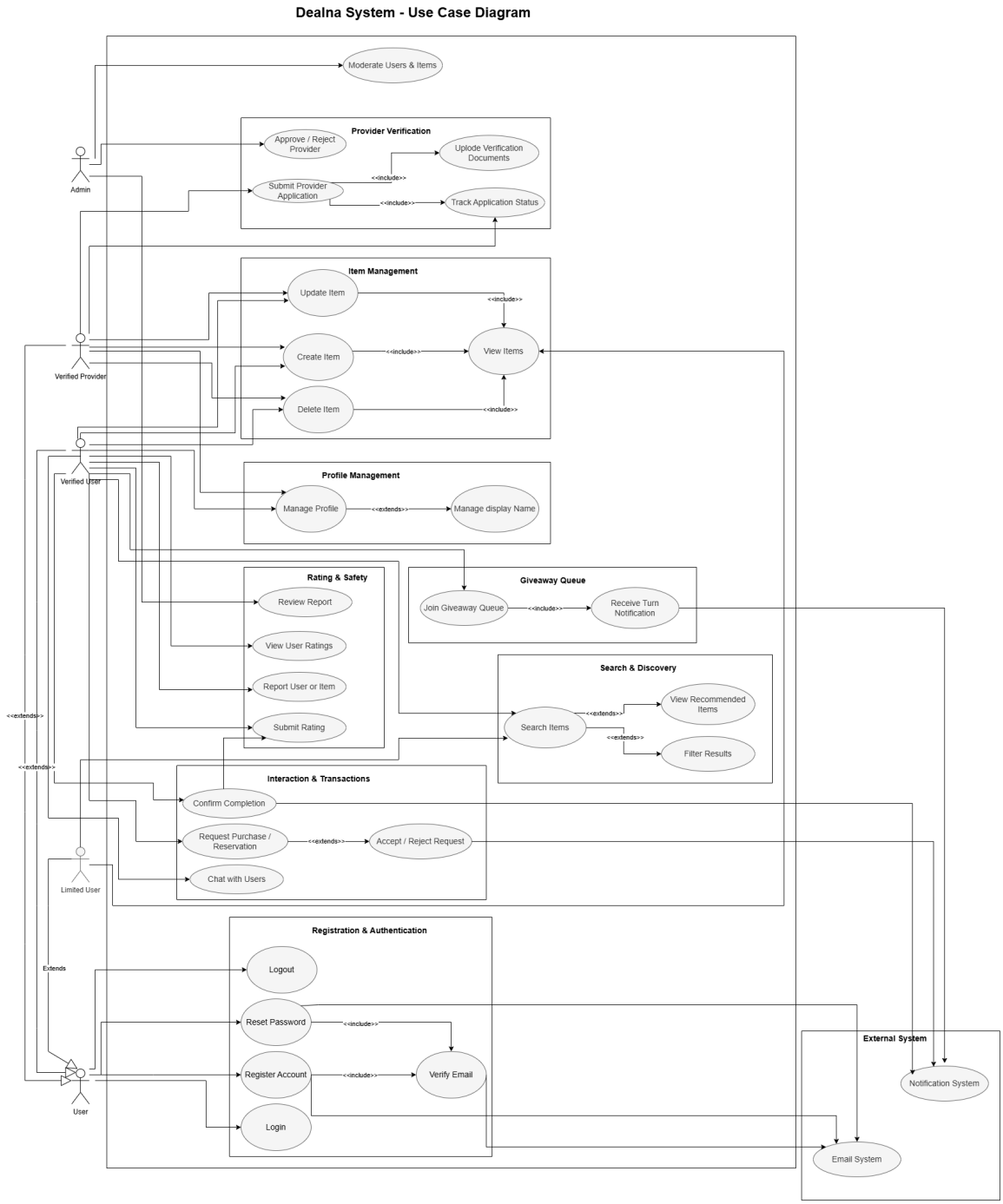

Figure 3.1: Dealna Use Case Diagram

Discussion:
The Dealna Use Case Diagram presents the system's main functionalities and how different actors interact with them. The User is the base actor, from which Limited User, Verified User, and Verified Provider inherit common actions such as registration and login, showing clear actor inheritance. Verified Providers can manage items (create, update, delete) through the Item Management Service, while Verified Users can rate, report, and join giveaway queues via the Rating & Safety Service and Giveaway Queue Service.

The Admin actor oversees system control by moderating users and items and approving or rejecting provider applications using the Administration Service. External systems such as the Email System and Notification System support verification and user notifications. An example of `«include»` is Register Account including Verify Email, ensuring mandatory verification within the Authentication Service. An example of `«extend»` is Request Purchase / Reservation extending to Accept / Reject Request in the Transaction Management Service, where the extension occurs only if a provider responds to the request.

3.2.4 Security Use Cases Diagram

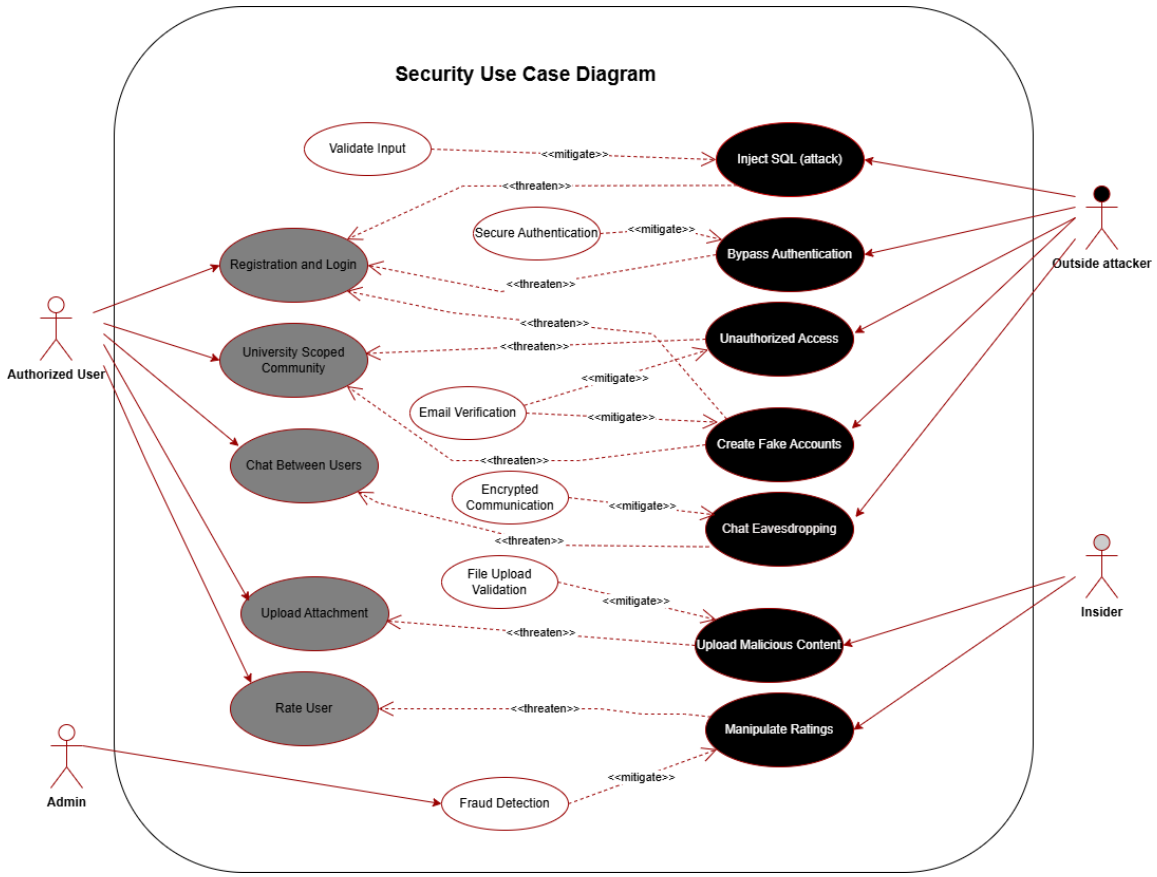

Figure 3.2: Dealna Security Use Case Diagram

Discussion:
This diagram shows the security critical use cases in our system and the main threats that may target them. It links each threat (SQL injection, unauthorized access, fake accounts, eavesdropping, malicious uploads, and rating manipulation) to the appropriate security controls we implemented, such as input validation, secure authentication (password hashing), email verification, encrypted communication (TLS+E2EE), file upload validation, and fraud detection with admin monitoring. Only the user requirements that directly require security protection are included in this diagram.

3.3 System Models

This section presents the main system models used to describe the structure and behavior of the Dealna platform. The diagrams collectively illustrate the static structure, dynamic interactions, workflows, and state transitions within the system.

3.3.1 Class Diagram

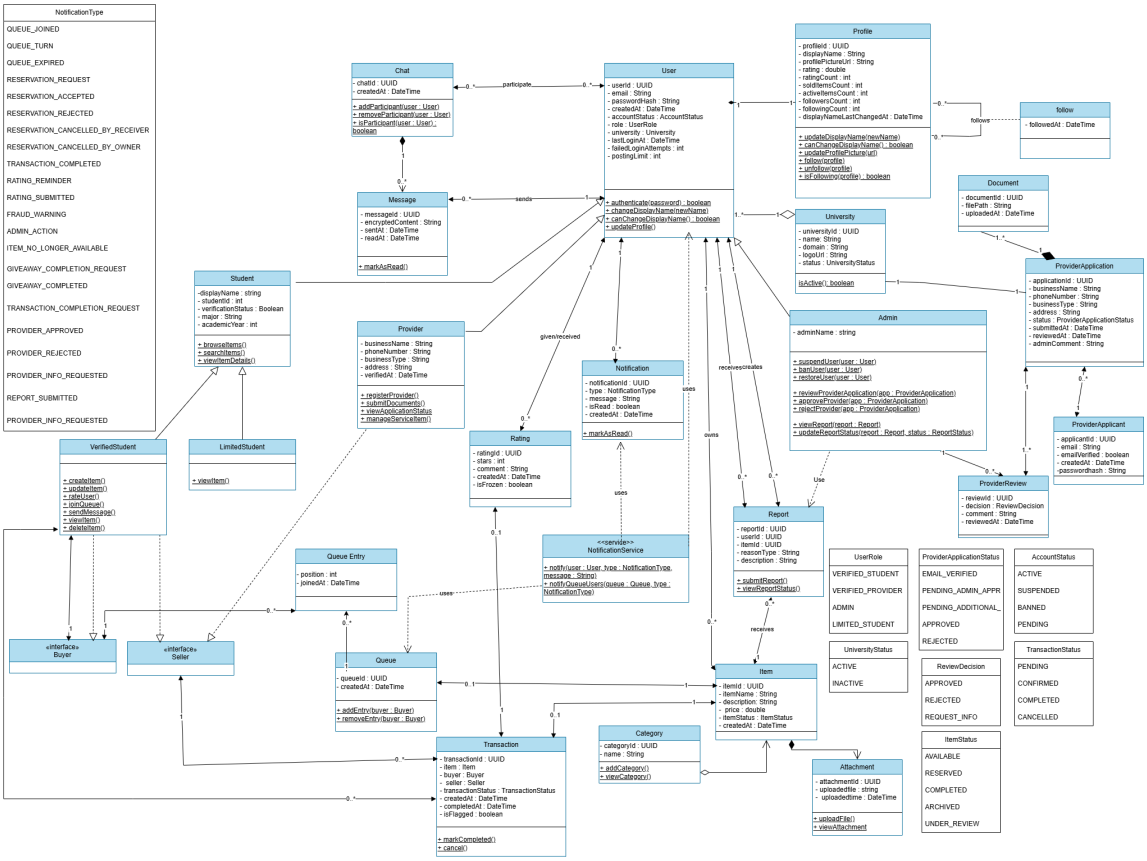

Figure 3.3: Dealna Class Diagram

Discussion:
The class diagram illustrates the static structure of the *Dealna* system and defines the relationships between its core entities. The *User* class acts as the central entity and is specialized into different roles through **inheritance**, enabling reuse of shared authentication and account attributes while separating role-specific responsibilities. The relationship between *User* and *Profile* is modeled as a **composition**, reflecting strong lifecycle dependency where a profile cannot exist independently of its user. Social interaction is represented through the *Follow* class, which models a **many-to-many self-association** between profiles. Marketplace items are linked to *Category* using **aggregation**, as categories exist independently of items, while *Attachment* is modeled as a **composition** due to its strict ownership by the item. Overall, the diagram demonstrates appropriate use of **inheritance, composition, aggregation, and association** to enforce business rules, maintain data integrity, and support a clear and maintainable system design.

3.3.2 Sequence Diagram

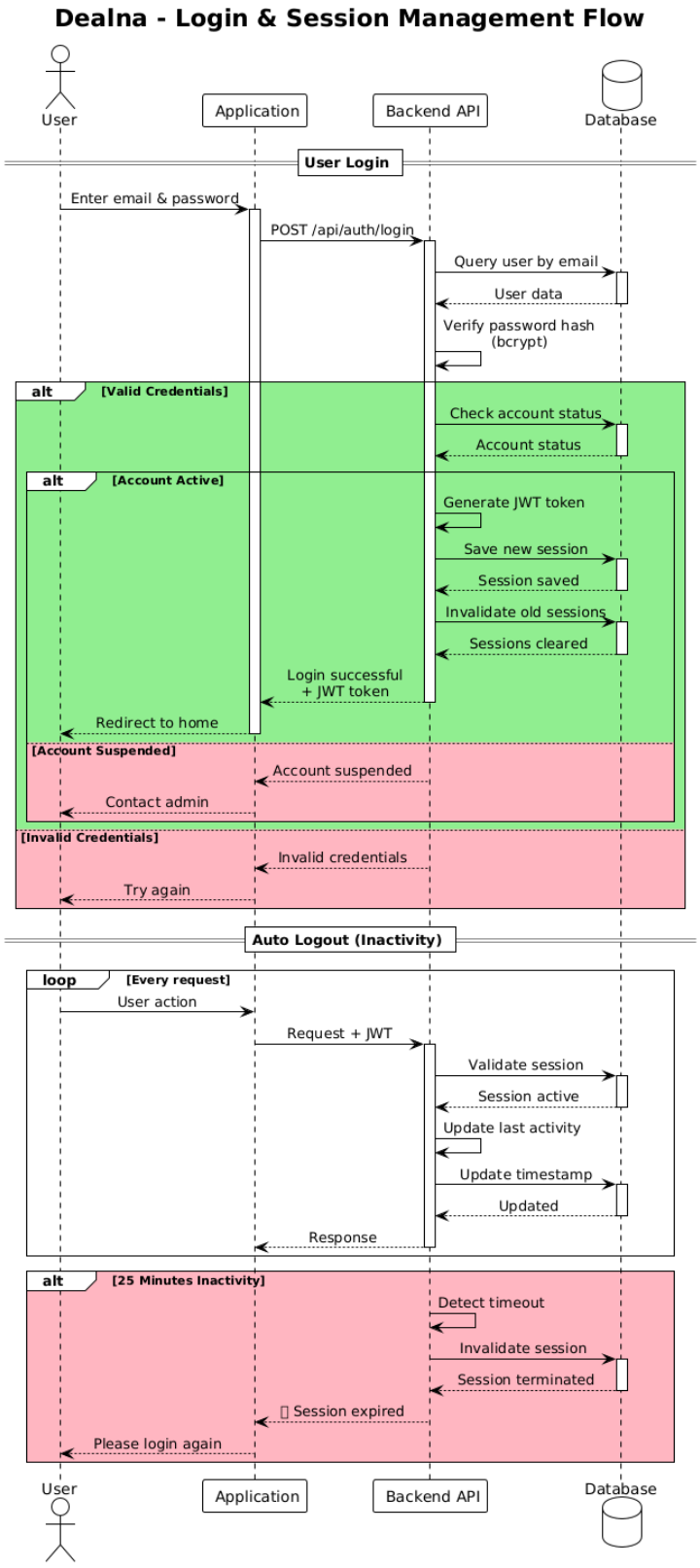

Figure 3.4: Login Session Sequence Diagram

Discussion:
This diagram illustrates the login flow and session management in the Dealna system: The user sends their credentials, and the system verifies them along with the account status. In case of success, a session is created with a JWT, while if the account is suspended or the credentials are incorrect, the user is returned to the login page. The system also manages sessions automatically, ending the session after 25 minutes of inactivity to ensure security.

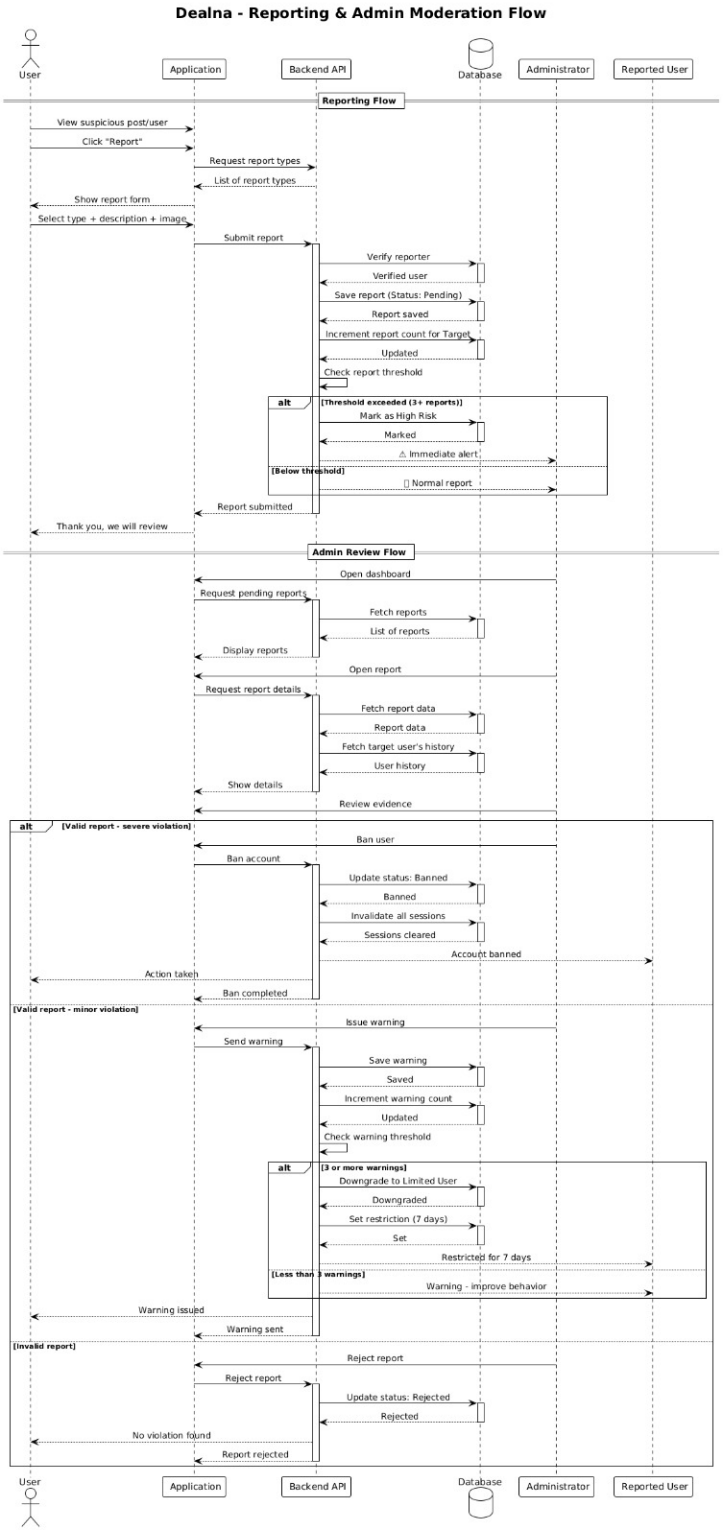

Figure 3.5: Report Admin Moderation Sequence Diagram

Discussion:

This diagram illustrates the reporting and administrative supervision system in Dealna: users can report suspicious content, and the reports are reviewed by moderators. In the case of a serious violation, the account is immediately suspended, while for minor violations, a warning is issued, and repeated warnings result in reduced privileges or a

temporary ban. The system also triggers an immediate alert when a certain threshold of reports is exceeded.

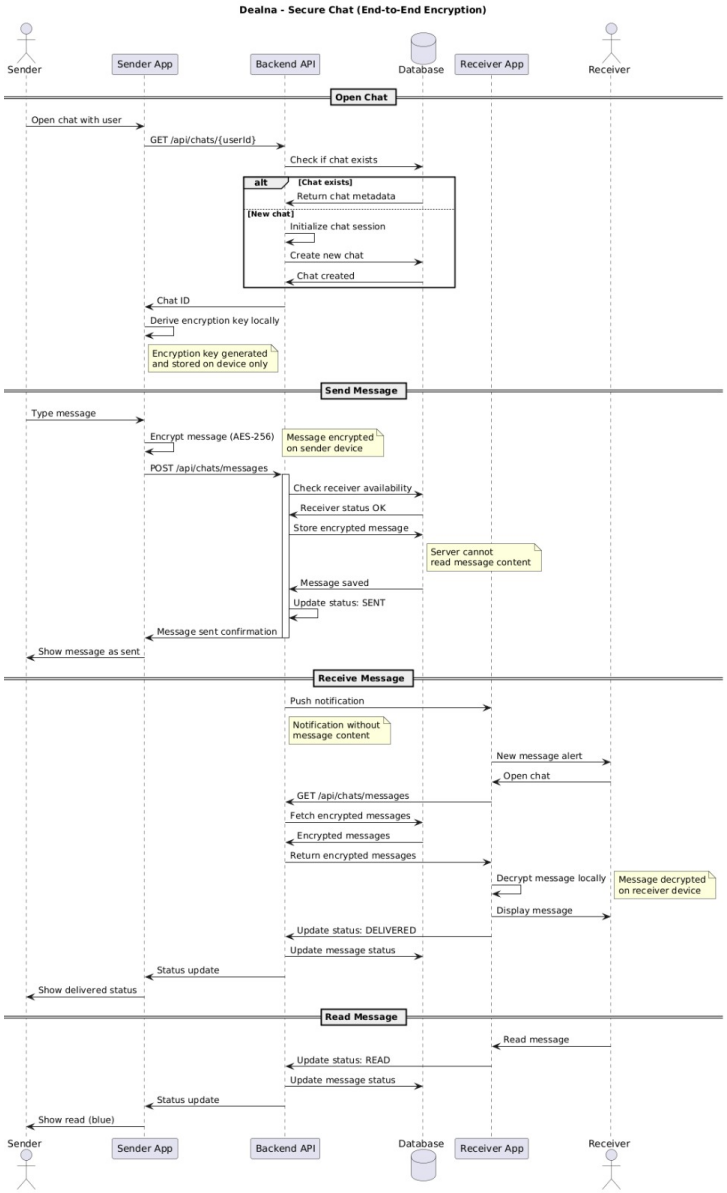

Figure 3.6: Secure Chat Sequence Diagram

Discussion:

Dealna – Secure Chat System Description: This diagram illustrates the end-to-end encrypted chat system in the Dealna app, involving the Sender App, Receiver App, Backend API, and Database. Messages are encrypted locally on the sender's device and decrypted only on the receiver's device, ensuring the server cannot access content. When a user opens a chat, the app retrieves or creates the chat via the Backend API and generates a local encryption key. Messages are encrypted before sending, stored encrypted in the database, and push notifications are sent without message content. Upon receiving, the app fetches and decrypts messages locally, updates delivery status, and notifies the sender when messages are read. The Backend API securely manages sessions, message storage, notifications, and status tracking but cannot decrypt messages. All encryption

keys remain on user devices, guaranteeing complete end-to-end security.

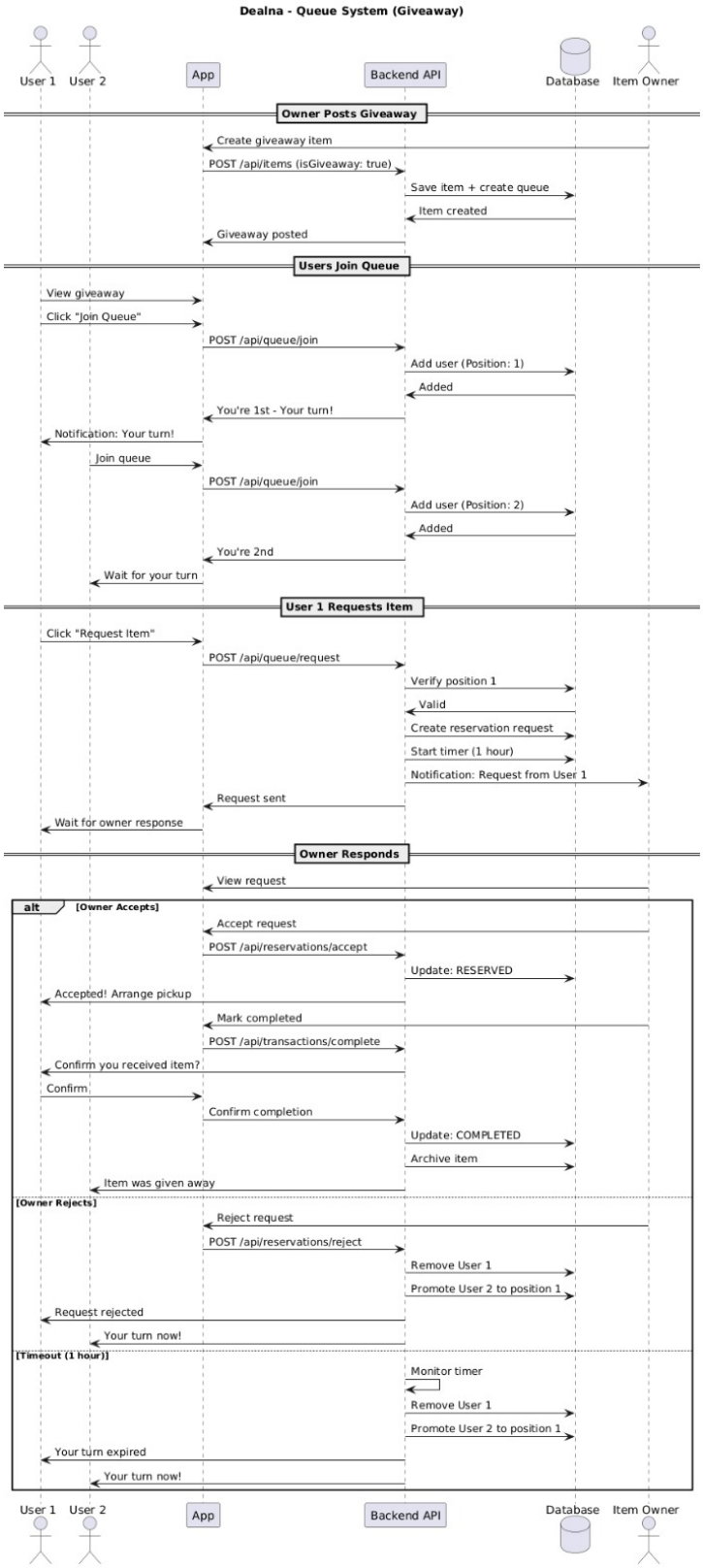

Figure 3.7: Queue System Sequence Diagram

Discussion:

This system regulates product donation processes through a waiting list: the owner posts the product, users join the list in order, and the first user requests the product. The owner then decides whether to accept or reject the request, and in case of rejection or expiration of time, the turn automatically moves to the next user.

3.3.3 Activity Diagram

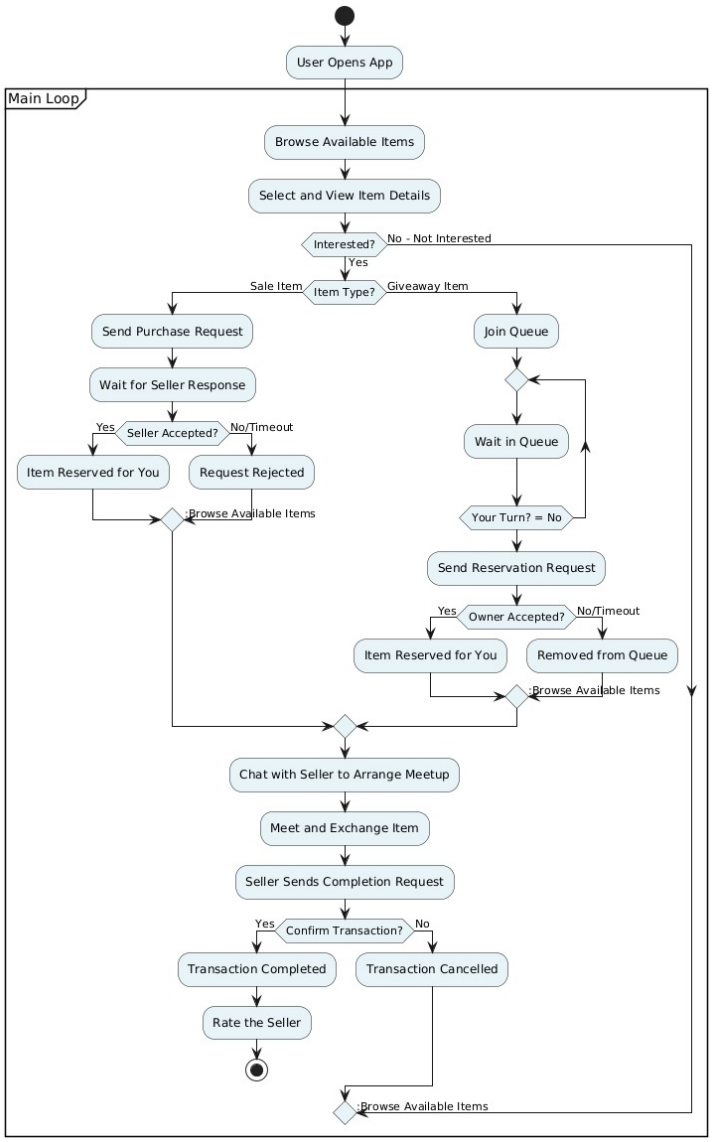

Figure 3.8: Activity Diagram

Discussion:
This activity diagram illustrates the complete flow of user interaction within the application. It shows the sequence of actions a user performs, starting from opening the app, browsing available items, viewing details, and deciding whether to proceed with a purchase or join a giveaway queue. Decision points, loops, and alternative paths are clearly represented, including how the system handles approvals, rejections, and transaction completion. This diagram helps reviewers understand the process logic and ensures that all user actions and system responses are covered.

3.3.4 State Chart Diagram

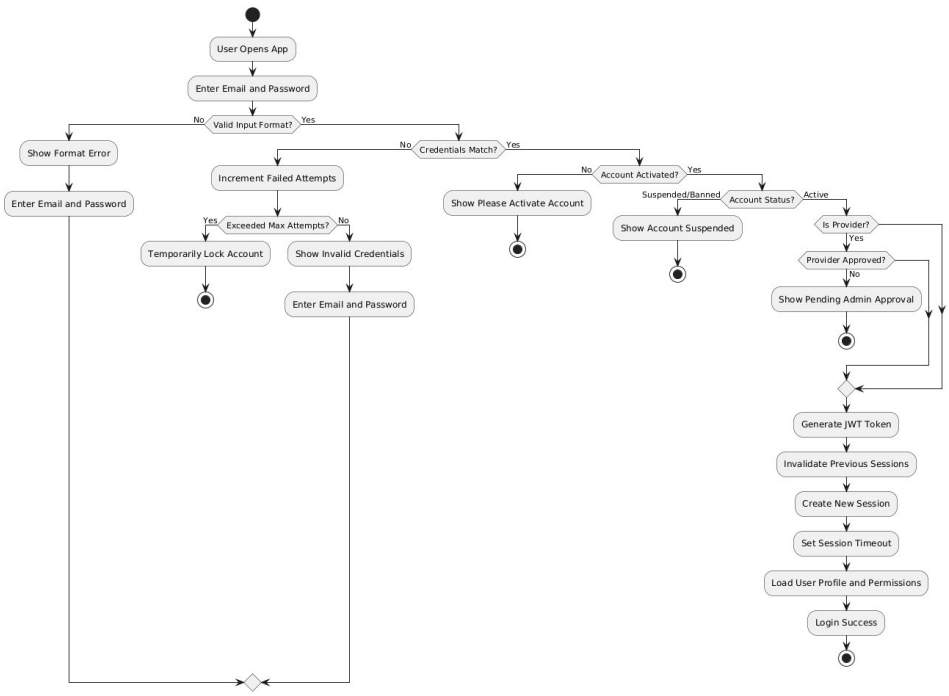

Figure 3.9: User Account State Chart Diagram

Discussion:
This diagram illustrates the secure login process: the user starts by entering their email and password, after which the data and account status (active, suspended, or requiring activation) are verified. If successful, a secure session is created with the appropriate permissions loaded, while error messages or security measures (such as locking the account after failed attempts) are displayed if a problem occurs.

3.4 System Architecture

3.4.1 Sub-System

1. **Account & Access Subsystem**
   - **Description:** This subsystem performs account lifecycle management, including registration, authentication, role assignment, and status control. It enforces access policies by validating credentials and mapping users to predefined permissions. Only entities with authorized roles are granted system access; all others are denied.
   - **Components:**
     - **User Registration Module**
     - **Authentication Module**
     - **Role & Permission Management Module**
     - **Account Status Management Module**
2. **Listings Subsystem**
   - **Description:** This subsystem responsible for usermanage listings.It allows people to post their products or services, organizes them into clear categories, and determines their availability. In short, it keeps the platform full of organized and easily accessible content.
   - **Components:**
     - **Listing Creation Module**
     - **Listing Management Module**
     - **Category Management Module**
     - **Item Status Management Module**
3. **Transaction Subsystem**
   - **Description:** This subsystem where the real work happen , any sale, purchase, or reservation between users goes through this subsystem. It's responsibility to tracks the process from start to finish or cancellation, ensuring every step is clear and properly monitored
   - **Components:**
     - **Reservation Module**
     - **Transaction Processing Module**
     - **Transaction Status Tracking Module**

4. **Communication Subsystem**
   - **Description:** This subsystem allows users to communicate with each other, whether through direct messages or system notifications. Its goal is to facilitate coordination, especially when there are listings or transactions. It makes communication easier and faster.
   - **Components:**
     - **Messaging Module**
     - **Notification Management Module**
5. **Trust & Safety Subsystem**
   - **Description:** This subsystem is the platform's guardian. It handles ratings, complaints, and monitoring to prevent any exploitation or fraud.His presence gives a sense of security and confidence, and makes people feel that they are dealing with a respectable and trustworthy place.
   - **Components:**
     - **Rating Management Module**
     - **Reporting Module**
     - **Moderation Module**
     - **Fraud & Abuse Prevention Module**

3.4.2 Software Architecture

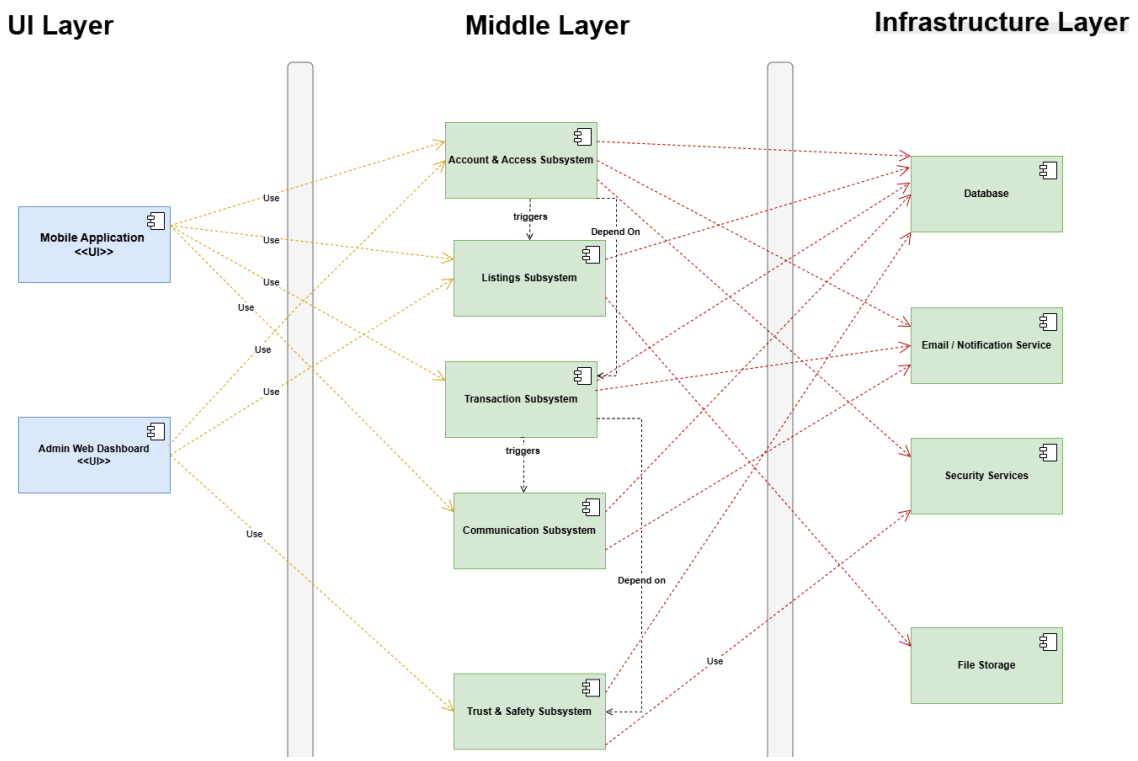

Figure 3.10: Software Architecture Diagram

Discussion:
This diagram presents the overall system architecture of Dealna, organized into three main layers: UI Layer, Middle Layer, and Infrastructure Layer. It shows how the user interfaces interact with the core subsystems, while all data access and security critical operations are handled through backend services. This layered design helps ensure separation of concerns, better maintainability, and improved system security.

3.4.3 Deployment Diagram

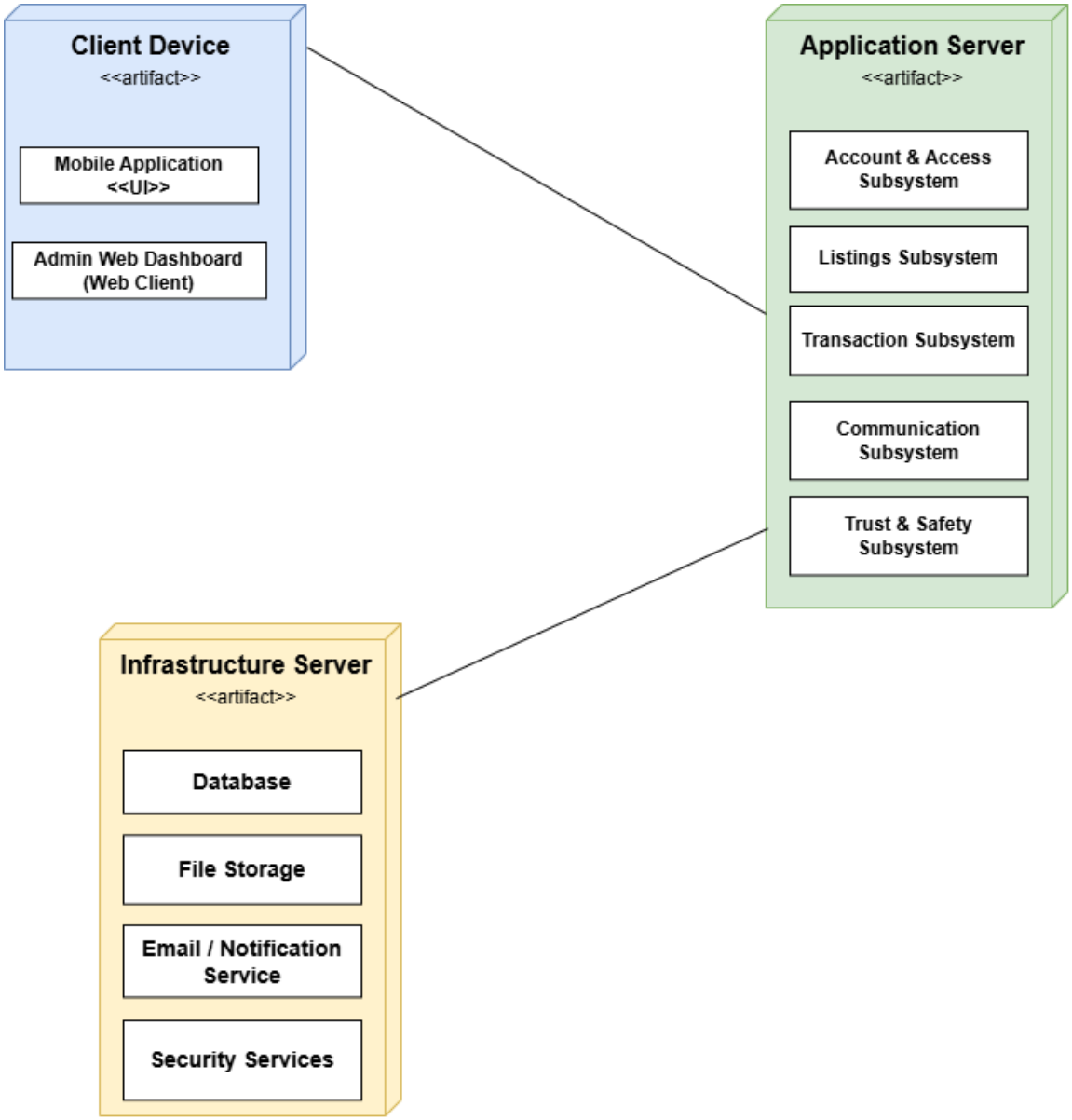

Figure 3.11: Deployment Diagram

Discussion:
This diagram illustrates the deployment structure of the Dealna system. It shows the distribution of system components across client devices, the application server, and the infrastructure server, and how these components interact to support system functionality.

3.5 Data Management and ERD

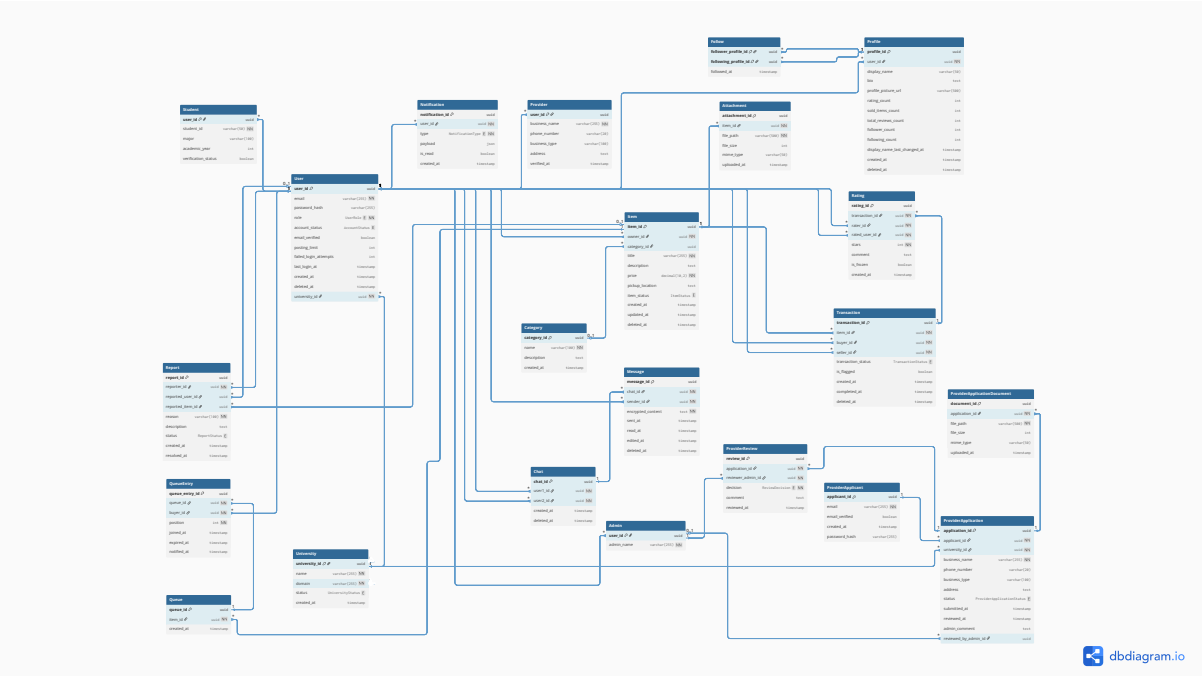

Figure 3.12: ERD Diagram

Discussion:
This ERD presents the logical data model of the Dealna system and how its core components interact. Each User is associated with exactly one University, enforcing university-scoped access. Users can own multiple Items, each classified under a single Category. A Queue with multiple QueueEntry records ensures fair competition among buyers for the same item. Transaction and Rating entities capture the exchange lifecycle and guarantee accountable and verifiable feedback.

Chapter 4

Conclusion and Future Works

Dealna addresses several limitations found in existing university marketplace platforms by providing a secure and controlled environment tailored specifically for students and verified service providers. By relying on university email verification, role-based access control, and structured transaction and reporting mechanisms, the system promotes trust and accountability among its users. Features such as queue-based giveaways, controlled rating aggregation, and secure one-to-one messaging further enhance the reliability of the platform. The proposed pilot deployment at Birzeit University offers an opportunity to validate the system design and evaluate its effectiveness in a real academic environment.

4.1 Review of the Project

This project focused on the design and implementation of *Dealna*, a university-oriented marketplace intended to support safe and regulated trading within the academic community. Unlike general-purpose marketplace applications, Dealna emphasizes identity verification and controlled access through distinct user roles, including verified students, limited students, verified service providers, and administrators.

The system architecture was designed with a clear separation of concerns between the backend and frontend components. The backend, implemented using Golang, is responsible for enforcing business logic, authentication, authorization, transaction rules, and database operations. In contrast, the Flutter mobile application focuses on user interaction, interface-level validation, and presenting data received from the backend without embedding core business rules.

Core functionalities such as user registration, login, item listing management, giveaway queues, transactions, ratings, reporting, and secure messaging were implemented based on a structured database schema and well-defined RESTful APIs. Additional attention was given to trust-related mechanisms, including rating stability, transaction abuse prevention, and administrative oversight. Overall, the project achieved its main objectives and demonstrates a practical application of full-stack development concepts within an academic context.

4.2 Next Semester Plan

In the next semester, the project will focus on completing, refining, and validating the existing implementation rather than introducing major architectural changes. The pri-

mary goal is to enhance system stability and prepare the platform for limited real-world use within a university setting.

The planned activities include:

- **Backend API Refinement and Layer Separation:** Improve the structure of Golang REST APIs by further separating controllers, services, and data access layers. Business rules and security logic will remain strictly on the backend to ensure maintainability and consistency.
- **Frontend–Backend Integration Stabilization:** Finalize communication between the Flutter application and backend APIs using consistent response formats. The Flutter client will rely entirely on backend decisions, reducing coupling and minimizing future modifications.
- **Authentication and Security Completion:** Complete authentication-related components such as session handling, token expiration, and email verification workflows, while preserving existing API contracts.
- **System Testing and Validation:** Conduct functional and integration testing for core use cases, including registration, item management, transactions, and secure messaging. Special focus will be given to edge cases and error handling.
- **Performance and Database Optimization:** Review database indexing strategies and API performance to ensure acceptable response times under typical student usage scenarios.
- **Pilot Deployment Preparation:** Prepare the system for a small-scale pilot deployment at Birzeit University, focusing on configuration, logging, and basic monitoring.

This phase will allow the project to mature from a functional prototype into a well-structured academic system that demonstrates sound software engineering practices and readiness for controlled deployment.

Bibliography

[1] G. Salton, A. Wong, and C. S. Yang, "A vector space model for automatic indexing," *Communications of the ACM*, vol. 18, no. 11, pp. 613–620, 1975.

[2] C. D. Manning, P. Raghavan, and H. Schütze, *Introduction to Information Retrieval*. Cambridge University Press, 2008.

[3] A. Huang, "Similarity measures for text document clustering," in *Proceedings of the Sixth New Zealand Computer Science Research Student Conference*, 2008.

[4] Baeldung, *Euclidean distance vs cosine similarity*, `https://www.baeldung.com/cs/euclidean-distance-vs-cosine-similarity`, 2024.

[5] OWASP Foundation, *Mobile top 10 – insecure communication*, `https://owasp.org/www-project-mobile-top-10/2016-risks/m3-insecure-communication`, 2023.

[6] B. H. Ibrahim et al., "Rule-based approach to e-commerce fraud detection," 2025.

[7] R. A. Damian, "Finding duplicate offers in the online marketplace," 2022.

[8] O. Albayrak, "Duplicate product record detection engine for e-commerce platforms," 2022.

[9] J.-L. Li, "E-commerce fraud detection model by computer ai," 2022.

[10] A. Mutemi and F. Bacao, "E-commerce fraud detection based on ml techniques," 2024.

[11] R. Sandhu et al., "Role-based access control models," 1996.

[12] OWASP Foundation, *Session management cheat sheet*, `https://owasp.org/www-project-cheat-sheets/cheatsheets/Session_Management_Cheat_Sheet.html`, 2023.

[13] M. Alnasser, *File upload security – essential practices for programmers*, 2024.

[14] Johannes Ullrich, *8 basic rules to implement secure file uploads*, `https://www.sans.org/blog/8-basic-rules-to-implement-secure-file-uploads`, Accessed: 2026-01-03, 2024.

[15] OWASP Foundation, *Sql injection prevention cheat sheet*, `https://cheatsheetseries.owasp.org/cheatsheets/SQL_Injection_Prevention_Cheat_Sheet.html`, 2023.

[16] *Yupa*, `https://www.yupa.co/`.

[17] *Uniexchange*, `https://uniexchange.app/`.

[18] *Daleel al talib*, `https://daleelaltalib.com/`.

[19] OWASP Foundation, *File upload cheat sheet*, OWASP Cheat Sheet Series.

[20] A. Jøsang, R. Ismail, and C. Boyd, "A survey of trust and reputation systems for online service provision," *Decision Support Systems*, vol. 43, no. 2, pp. 618–644, 2007.

[21] P. Resnick, R. Zeckhauser, E. Friedman, and K. Kuwabara, "Reputation systems," *Communications of the ACM*, vol. 43, no. 12, pp. 45–48, 2000.

[22] J. L. Herlocker, J. A. Konstan, L. G. Terveen, and J. T. Riedl, "Evaluating collaborative filtering recommender systems," *ACM Transactions on Information Systems*, vol. 22, no. 1, pp. 5–53, 2004.

[23] Y. Wang and M. P. Singh, "Formal trust model for multiagent systems," in *Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI)*, 2014.

Appendices

.1 Rating Algorithms Comparison and Selection

Online reputation systems can rely on different rating aggregation algorithms. Each approach has distinct strengths and limitations, especially in environments with varying numbers of ratings. This appendix compares common rating algorithms and justifies the selection of the Bayesian average for the proposed system [20], [21].

.1.1 Simple Average Rating

The simplest rating approach calculates the arithmetic mean of all received ratings:

$$R = \frac{\sum r_i}{n}$$

Limitations:

- Highly sensitive to early ratings.
- A small number of extreme ratings can significantly distort the result.
- Does not reflect confidence or reliability when the number of ratings is low.

This approach is unsuitable for trust-based systems where fairness and stability are required [21].

.1.2 Weighted Average Rating

Weighted average methods assign different weights to ratings based on predefined rules:

$$R = \frac{\sum(r_i \times w_i)}{\sum w_i}$$

Limitations:

- Requires manual tuning of weights.
- Weight rules may be arbitrary or inconsistent across scenarios.
- Less effective when user behavior changes over time.

Although this method improves over the simple average, it still lacks a principled probabilistic foundation [22].

.1.3 Bayesian Average Rating (Selected Approach)

The proposed system adopts a Bayesian average rating model:

$$R = \frac{n \cdot \bar{r} + m \cdot C}{n + m}$$

where:

- $\bar{r}$ is the user's average rating.
- n is the number of received ratings.
- C is the global platform average.
- m is a smoothing constant.

Advantages:

- Prevents extreme ratings when the number of raters is small.
- Gradually shifts from prior belief to actual user behavior.
- Automatically adjusts confidence without manual weighting rules.
- Widely used in reputation and trust systems [20], [22].

.1.4 Algorithm Selection Justification

The Bayesian average was selected as it provides the best balance between fairness, stability, and implementation simplicity. Unlike simple or manually weighted averages, the Bayesian approach incorporates uncertainty handling through probabilistic smoothing, making it well-suited for student marketplace environments where users may initially have limited transaction history.

This method ensures that:

- New users are not unfairly advantaged or penalized.
- Ratings become increasingly reliable as evidence accumulates.
- The system remains resistant to manipulation and biased behavior.

Such properties are strongly recommended in trust-based and reputation-driven platforms [20], [21], [23].

.2 Use Case Specifications

.2.1 Report Content

Title	Report Content
Description	This use case explains how a verified student can report an item or inappropriate user behavior within the application.
Actor	Verified Student
Pre-condition	• The user must be logged into the system. • The user's role must be a verified student and not a limited student.
Main Flow	1. The user views an item, user profile . 2. The user selects the report option available in the application interface. 3. The system displays a report submission form that includes report type selection and an optional text description. 4. The user selects a report type such as fraud, duplicate item, scam, harassment, or inappropriate content. 5. The user optionally enters a textual description. 6. The user optionally uploads an attachment such as a screenshot of a chat message that related to the report. 7. The user submits the report. 8. The system stores the report with the reporting user ID, reported entity ID, timestamp, and attachment (if provided). 9. The system sets the report status to pending review. 10. The system notifies the administrator about the submitted report.
Alternative Flow (A)	**Missing Required Field:** A1. The system displays an error message indicating that the report type is required. A2. The user selects a valid report type. A3. The user resubmits the report form.
Alternative Flow (B)	**Attachment Too Large:** B1. The system rejects the file upload. B2. The system displays an error message indicating that the uploaded file size exceeds the allowed limit of 3 MB .
Post-condition	The report is securely stored in the system database. The report becomes visible on the administrator dashboard for review.

Table 1: Use Case Description – Report Content

.2.2 Service Provider Registration

Title	Service Provider Registration
Description	This use case explains how the system handles the registration process of an external service provider, including email verification, document submission, and administrative approval.
Actor	Service Provider, Administrator
Pre-condition	The service provider does not need to have an existing account. The service provider must have a valid email address. The system must be operational to allow provider registration and admin review.
Main Flow	1. The service provider selects *Register as Service Provider*. 2. The system requires the provider to enter an email address and a password. 3. The provider submits a valid email address & password. 4. The system verifies that the email address is not already registered and validates the password according to predefined security rules . 5. The system sends an activation link to the provider's email address. 6. The provider clicks the activation link. 7. The system verifies the link and marks the email as verified. 8. The system redirects the provider to the registration form. 9. The provider submits required information including business details and legal documents. 10. The system validates the submitted information and prevents duplicate accounts based on email address and phone number. 11. The system sets the account status to *Pending Admin Approval*. 12. The administrator reviews the application. 13. The administrator approves the application. 14. The system upgrades the account status to *Verified Provider*. 15. The system sends an approval notification to the provider. 16. The provider gains access to the platform and posting features.
Alternative Flow (A)	**Email Already Exists:** A1. The system displays an error message indicating the email is already registered.
Alternative Flow (B)	**Activation Link Not Clicked:** B1. The account remains in *Awaiting Email Verification* state.
Alternative Flow (C)	**Activation Link Expired:** C1. The activation link expires before the service provider clicks it. C2. The system displays a message show that the activation link has expired. C3. The service provider requests a new activation link. C4. The system resends a new activation link, subject to resend attempt limitations.
Alternative Flow (D)	**Admin Rejects Application:** D1. The system sets the account status to *Rejected*. D2. A rejection email with the reason is sent to the provider.
Alternative Flow (E)	**Admin Requests Additional Information:** E1. The system updates the account status to *Additional Information Required*. E2. The provider uploads additional documents. E3. The application status returns to *Pending Admin Approval*.
Post-condition	**Success:** The provider account is verified and granted posting access. **Failure:** The provider application is rejected and the provider is notified with the rejection reason.

Table 2: Use Case Description – Provider Registration

.2.3 Secure Chat Between Users

Title	Secure chat between users
Description	The system provides a secure platform for different user roles to access and interact with available items. Users log in using their email and password, and access is granted based on their role and account status. Students can browse items and navigate through categories or recent lists, as well as search for specific items easily. The system is designed to be simple, secure, and user-friendly.
Actor	Verified Student , verified provider
Pre-condition	1. The user logs into the system . 2. The user is neither banned nor suspended 3. The other party is the same 4. There is communication between the two parties (e.g., inquiry about an advertisement or making a purchase).
Main Flow	1. The user opens the application. 2. The user selects an old or new conversation with another user 3. The RBAC system is used to verify user permissions. 4. The user can write a message or send a picture or attachment 5. Client-side Encryption is performed on the user's device 6. The message is sent to the server in encrypted form. 7. The server stores the message without being able to read its contents. 8. The other party receives the message 9. The message status is displayed as either (Sent / Delivered / Read) by the system 10. The other party reads the message in the same conversation
Alternative Flow (A)	**The user account is suspended or banned:** A1. In case the user is banned or suspended . A2. Sending messages by the system is prohibited . A3. A clarification message appears to the user . A4. The process ends.
Alternative Flow (B)	**Attachment Not Allowed:** B1. If the size of the attachment uploaded by the user exceeds the allowed limit or is unsupported: B2. The system refuses to send. B3. An error message is displayed to the user B4. The user returns to the message input step
Alternative Flow (C)	**Chat Encryption Failure:** C1. If encryption fails on the user's device: C2. The message is not being sent. C3. The system displays a failure message. C4. The process is finished.
Post-condition	1. The message has been sent successfully . 2. The message is encrypted and stored . 3. The status of messages is tracked and updated according to the stages: (sent, received, and read) .

Table 3: Use Case Description – Secure chat between users

.2.4 Manage Item Listing

Title	Manage Item Listing
Description	This use case allows verified students and verified providers to manage item listings within the system. Users can create new items, edit existing ones, or delete items they no longer wish to offer. The system validates all entered data and uploaded images before saving the item and making it visible in active lists according to user roles and market rules.
Actor	Verified Student , verified provider
Pre-condition	1. 1. The user logs in . 2. The user is neither banned nor suspended 3. The user can create or edit an item .
Main Flow	1. The user opens the application. 2. The user selects 'Create Item' or opens an existing item to manage it . 3. The user enters the required information: title, description, category, condition, price (if any), and pickup location . 4. The user uploads 1–5 images of the item. 5. The system verifies the validity of the entered data (text, image formats, size, and number of images). 6. The system saves the item and displays it in the user's active lists. 7. The user can later edit or delete the item: • Edit: Change any allowed field or replace images. • Delete: Remove the item from active lists.
Alternative Flow (A)	**Invalid Input:** A1. If the system detects incorrect texts or unsupported image formats : A2. The system displays an error message . A3. The user must correct the data .
Alternative Flow (B)	**Image Limit Exceeded:** B1. If the user uploads more than 5 images : B2. The system prevents submission . B3. The system displays an error message indicating that only 5 images are allowed and requests removal of the extra images .
Post-condition	1. The item has been successfully created, modified, or deleted . 2. The current version of the item and its associated images have been stored in the database . 3. Items appear in active lists according to the user's role and market rules .

Table 4: Use Case Description – Manage Item Listing

.3 Transaction Gaming Prevention and Anti-Fraud Logic

Online reputation systems are susceptible to manipulation through coordinated or repetitive transactions between the same users. Such behavior may be exploited to artificially inflate reputation scores without representing genuine marketplace activity. This appendix outlines the threshold-based and history-aware mechanisms employed by the system to prevent transaction gaming and preserve rating integrity.

- **Pairwise Transaction History Monitoring.** The system maintains a historical interaction record for each unique buyer–seller pair. This record includes completed transactions, associated ratings, transaction timestamps, and rating values. Historical data is evaluated over rolling time windows to distinguish legitimate repeated interactions from potentially manipulative behavior.

- **Baseline Threshold Definition.** A baseline threshold of approximately five completed transactions between the same pair of users within a rolling 30-day window is applied. This value is selected to balance the detection of coordinated rating manipulation while allowing legitimate repeated interactions. Transactions and ratings

that exceed this threshold are considered potentially suspicious and are subject to further evaluation.

- **Adaptive Threshold Adjustment.** Threshold values are not strictly fixed and may be adaptively adjusted based on historical interaction patterns, transaction timing, and rating behavior. For example, transactions that are evenly distributed over time or exhibit diverse rating values may be treated as less suspicious than rapid, repetitive transactions with consistently maximum ratings.

- **Suspicious Pattern Detection.** When the number of completed transactions between a user pair exceeds the defined threshold, the system evaluates additional indicators, including transaction frequency, rating symmetry, and rating extremity. If suspicious patterns persist, subsequent ratings may be flagged and excluded from public rating calculations.

- **Administrative Notification.** Exceeding predefined thresholds automatically triggers an administrative notification. Administrators are provided with relevant contextual information, including transaction history, rating timelines, and threshold exceedance details, to support informed review decisions.

- **Temporary Rating Freeze.** In cases of elevated suspicion, affected ratings may be temporarily frozen. Frozen ratings are not included in reputation calculations until an administrative review is completed, preventing immediate exploitation while preserving the possibility of reinstatement.

- **Administrative Review and Enforcement.** Administrators may review flagged cases and apply appropriate actions, including approving ratings, permanently excluding them from calculations, issuing warnings, or applying account-level restrictions. This layered approach combines automated detection with human oversight to ensure fairness, accuracy, and system integrity.

Overall, the proposed mechanism provides a balanced and scalable approach to transaction gaming prevention, minimizing false positives while effectively protecting the credibility of the system's reputation framework.

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